SCIE 1120 Chemistry and Life

生物合成

Protein Biosynthesis

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LT-J

The Flow of Genetic Information

- The information content of DNA is in the form of specific **sequences of nucleotides**
- The DNA inherited by an organism leads to specific traits by dictating the synthesis of proteins 性状
- Proteins are the links between **genotype** 基因型 (by observing DNA) and **phenotype** 表型 (Observable characteristics at microbe (single cell organism)/organismal levels)
- **Gene expression**, the process by which DNA directs protein synthesis, includes two stages: **transcription** and **translation**
- Some genes only express a given phenotype in certain environmental conditions. Conversely, some phenotypes could be the result of multiple genotypes.

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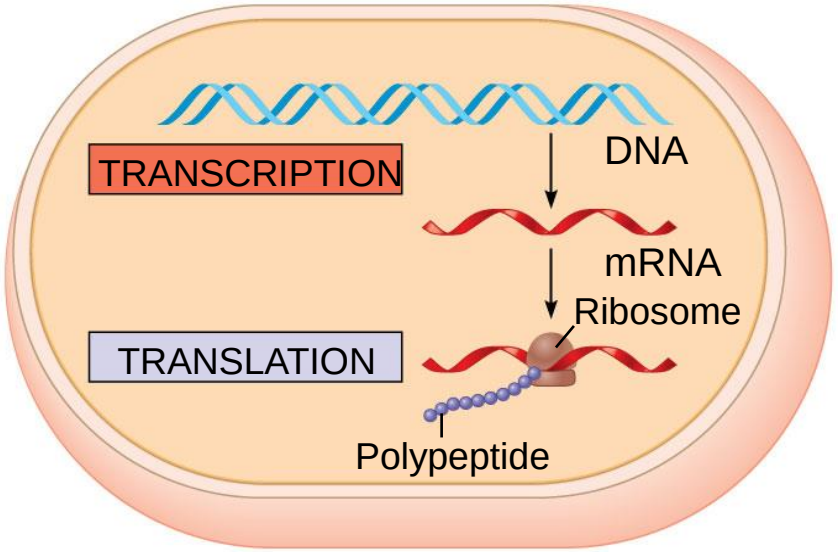
Basic Principles of Transcription and Translation

- RNA is the bridge between genes and the proteins for which they code
- **Transcription** is the synthesis of **messenger** RNA using information in DNA
- **Translation** is the synthesis of a polypeptide, using information in the mRNA
- **Ribosomes** are the sites of translation

转录是以 DNA **中的信息合成信使** RNA 的过程。
翻译是利用信使核糖核酸（mRNA）**中的信息合成**多肽的过程。
核糖体是蛋白质合成的场所

中心法则

The central dogma is the concept that cells are governed by a cellular chain of command



(a) Bacterial (prokaryotic) cell

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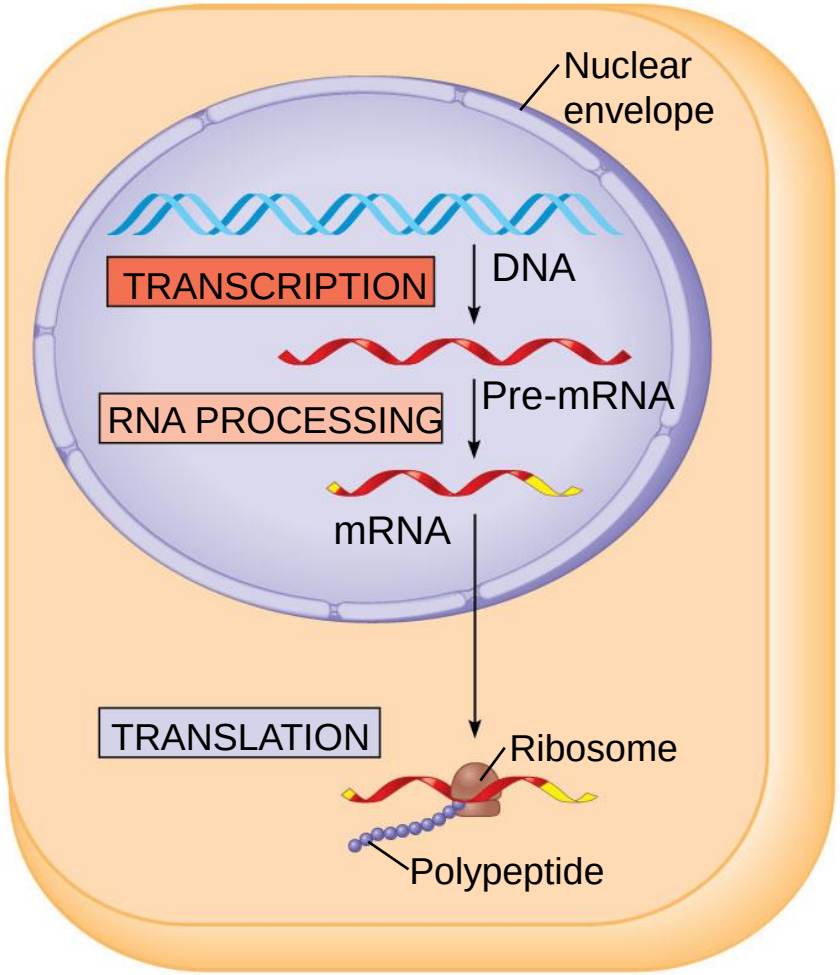

(b) Eukaryotic cell

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Four Levels of Protein Structure

- The primary structure of a protein is its unique sequence of amino acids (20 aa, determines the 3D structure of a protein)
- Secondary structure, found in most proteins, consists of coils and folds in the polypeptide chain 多肽链的螺旋和折叠
- Tertiary structure is determined by interactions among various side chains (R groups) 侧链的相互作用
- Quaternary structure results when a protein consists of multiple polypeptide chains 多条肽链

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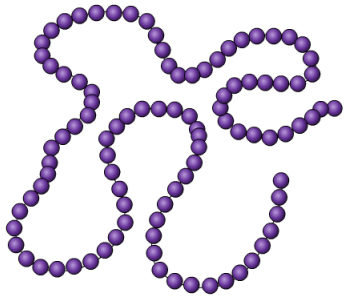
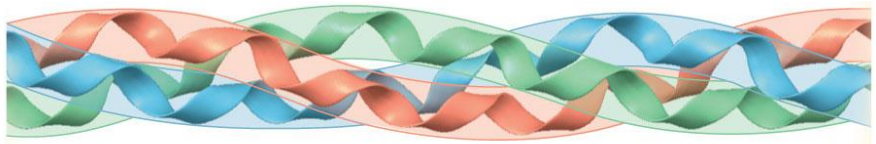
Figure 5.20b



Primary structure

Collagen



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Secondary structure

Tertiary structure

Quaternary structure

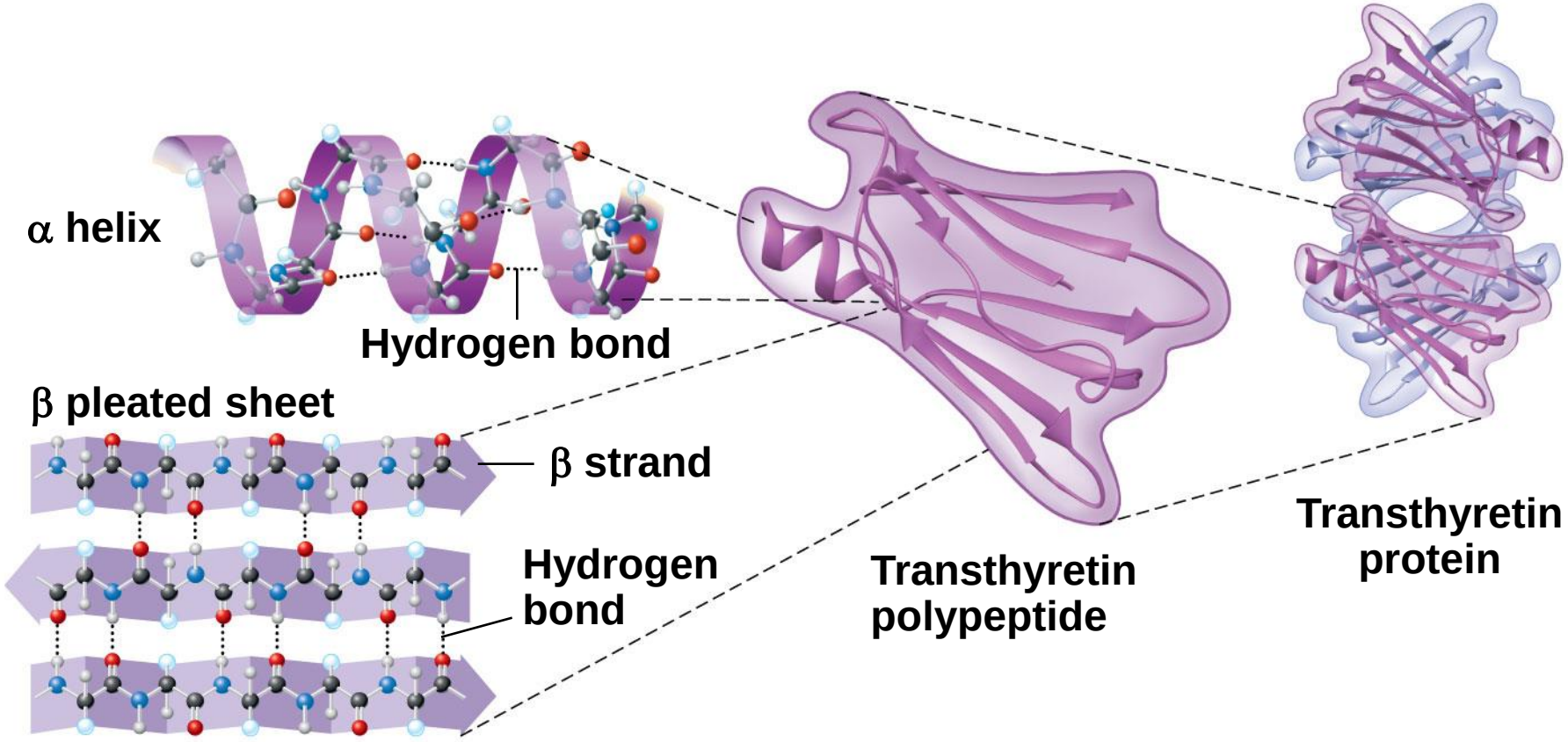

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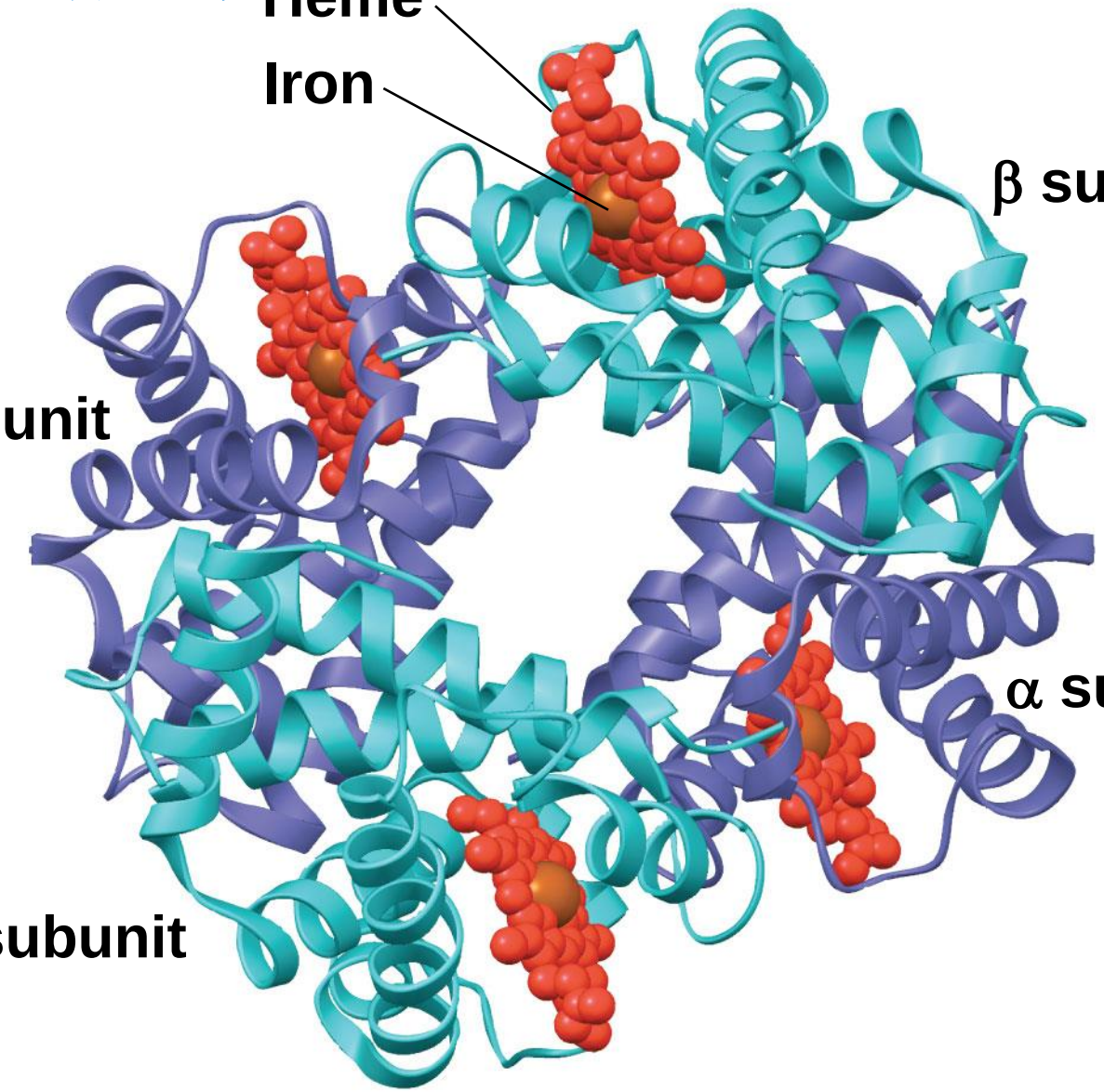
Figure 5.20i



Hemoglobin 血红蛋白

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蛋白质包含多种结构，因而具有广泛的功能。

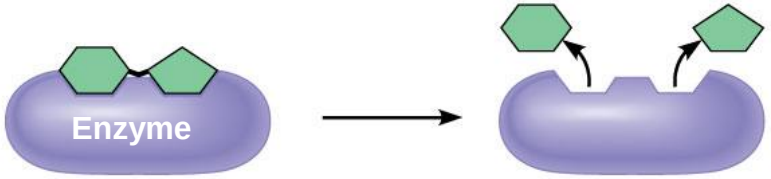
Proteins include a diversity of structures, resulting in a wide range of functions

- Proteins account for more than 50% of the dry mass of most cells
- Protein functions include structural support, storage, transport, cellular communications, movement, and defense against foreign substances

Enzymatic proteins 酶蛋白

Function: Selective acceleration of chemical reactions

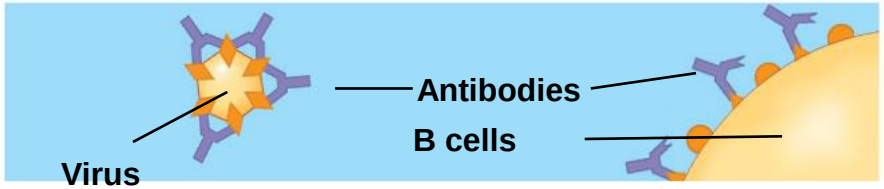
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins 防御蛋白

Function: Protection against disease

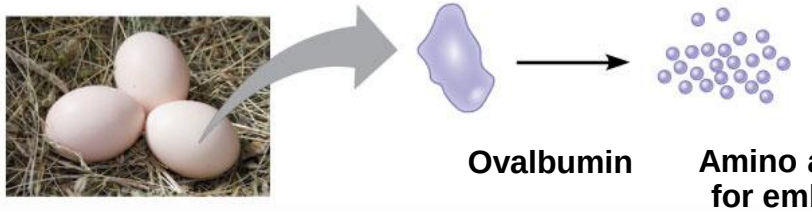
Example: Antibodies inactivate and help destroy viruses and bacteria. 抗体



Storage proteins 贮藏蛋白

Function: Storage of amino acids

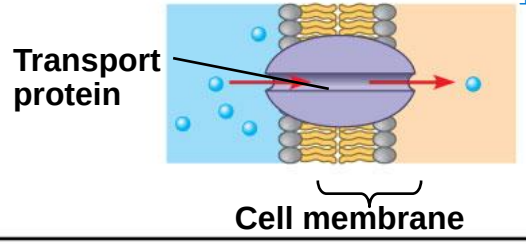
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins 转运蛋白

Function: Transport of substances

Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across cell membranes. 血红蛋白



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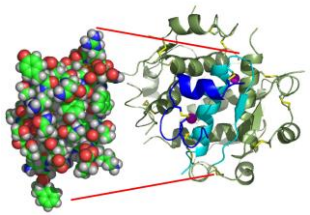
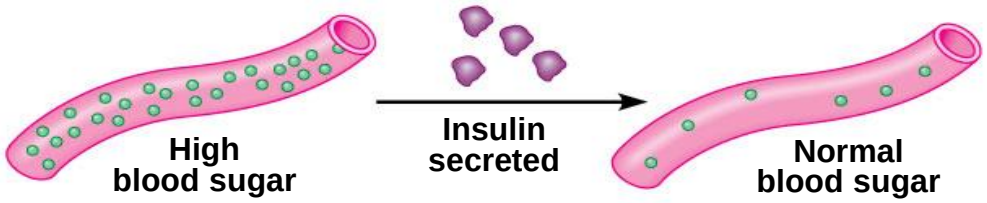
Hormonal proteins 激素蛋白

Function: Coordination of an organism's activities

Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration

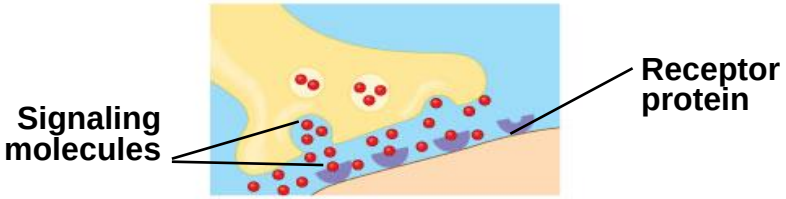




Receptor proteins

Function: Response of cell to chemical or biological stimuli

Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.

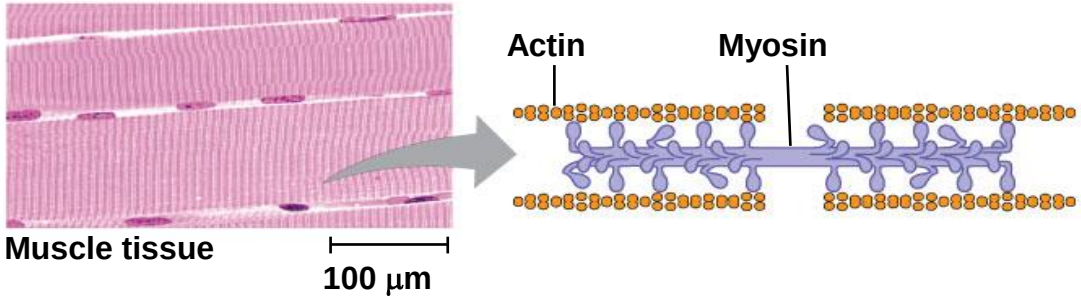


Contractile and motor proteins 收缩蛋白和运动蛋白

Function: Movement

Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.

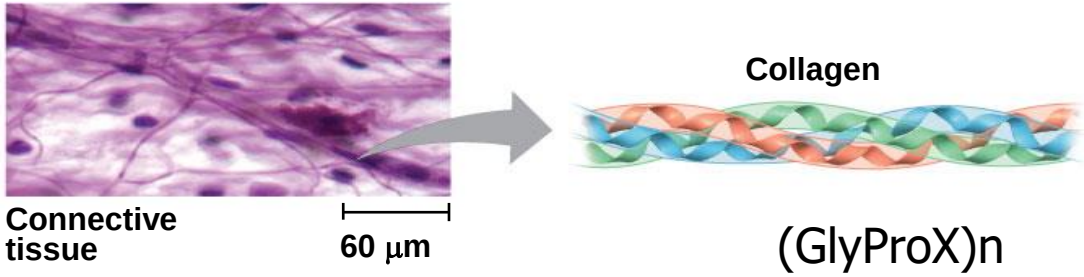
马达蛋白/肌动蛋白



Structural proteins 结构蛋白

Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



(GlyProX)n

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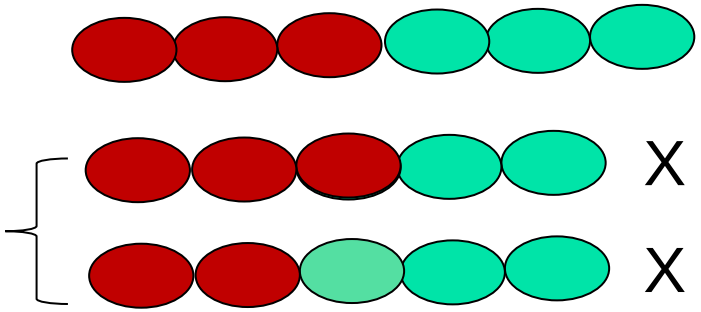
密码子：核苷酸三联体

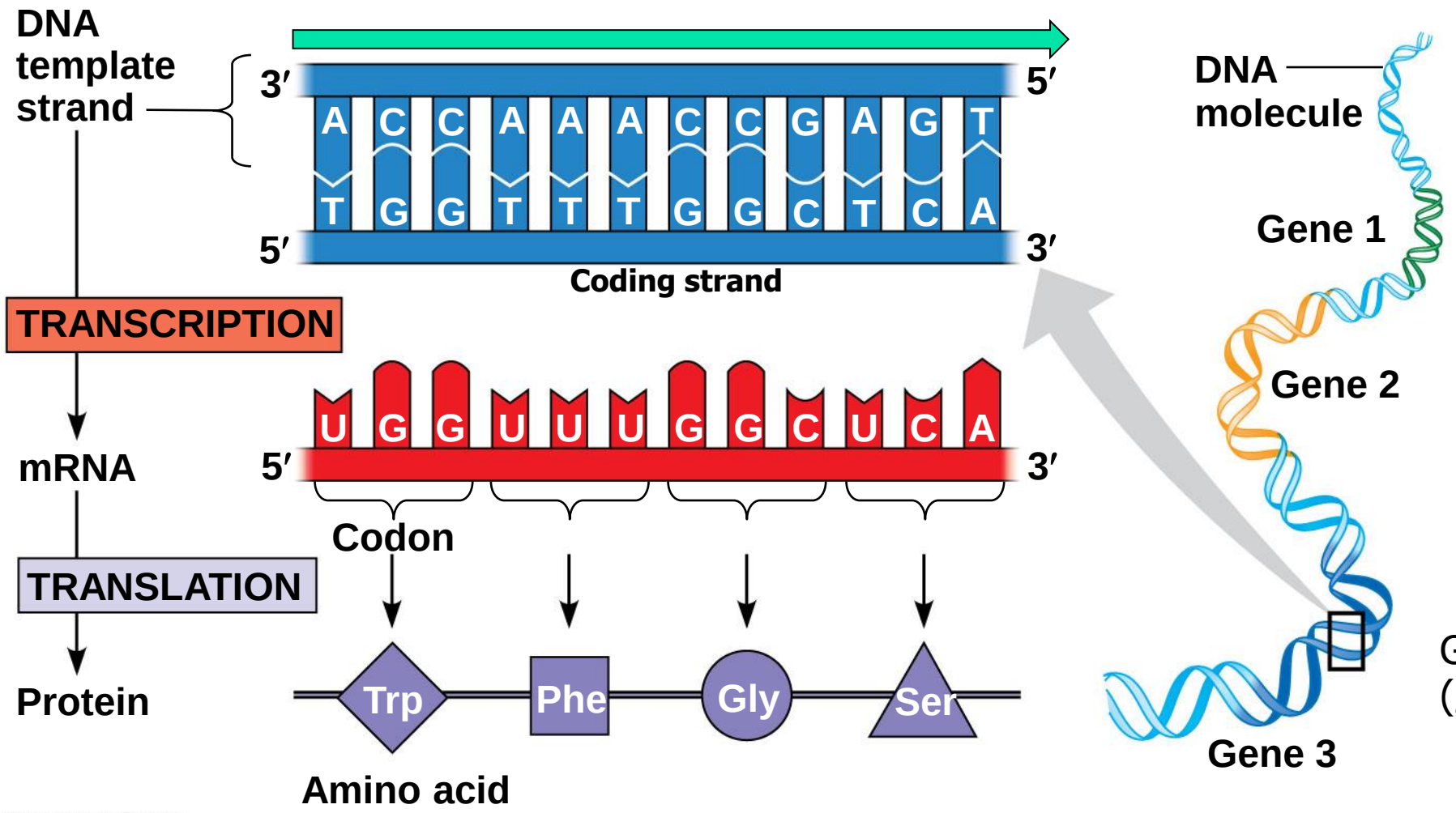
Codons: Triplets of Nucleotides



- The flow of information from gene to protein is based on a **triplet code**: a series of **non-overlapping**, three-nucleotide words
- The words of a gene are transcribed into complementary nonoverlapping three-nucleotide words of mRNA (each base can only be read once)
- These words are then translated into a chain of amino acids, forming a polypeptide

Singlet and doublet (4, 16) are not enough for 20 amino acids. Triplet (64) is enough for 20 amino acid with some redundancy. Quadruplet (256) is too many for the 20 amino acids, it will not be efficient for cell .

Codons: Triplets of Nucleotides



Gene Activation
(genes "on")

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Codons: Triplets of Nucleotides

- During transcription, one of the two DNA strands, called the **template strand**, provides a template for ordering the sequence of complementary nucleotides in an RNA transcript
- During translation, the mRNA base triplets, called **codons**, are read in the 5′ to 3′ direction
- Codons along an mRNA molecule are read by translation machinery in the **5′ to 3′** direction
- **Each codon** specifies the amino acid (one of 20 aa) to be placed at the corresponding position along a polypeptide

Cracking the Code

- All 64 codons were deciphered by the mid-1960s
- Of the 64 triplets, 61 code for amino acids (including one for "start"); 3 triplets are "stop" signals to end translation
- The genetic code is redundant (more than one codon may specify a particular amino acid) but not ambiguous; **no codon specifies more than one amino acid**

Second mRNA base

First mRNA base (5′ end of codon)	U	C	A	G	Third mRNA base (3′ end of codon)
U	UUU, UUC Phe	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr	UGU, UGC Cys	U
	UUA, UUG Leu		UAA Stop	UGA Stop	C
			UAG Stop	UGG Trp	A
					G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His	CGU, CGC, CGA, CGG Arg	U
			CAA, CAG Gln		C
					A
					G
A	AUU, AUC, AUA Ile	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn	AGU, AGC Ser	U
	AUG Met or start		AAA, AAG Lys	AGA, AGG Arg	C
					A
					G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp	GGU, GGC, GGA, GGG Gly	U
			GAA, GAG Glu		C
					A
					G

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Cracking the Code

- Codons must be read in the correct **reading frame** (correct groupings) in order for the specified polypeptide to be produced
- An **open reading frame (ORF)** is a reading frame that contains a **start codon**, and a subsequent region which usually has a length which is a multiple of 3 nucleotides but does not contain a **stop codon** in a given reading frame. An **open reading frame starts** with an atg (Met) in most species and **ends** with a **stop codon** (taa, tag or tga)

Second mRNA base

First mRNA base (5′ end of codon)	U	C	A	G	Third mRNA base (3′ end of codon)
U	UUU Phe UUC	UCU UCC UCA UCG Ser	UAU Tyr UAC	UGU Cys UGC	U
	UUA Leu UUG		UAA Stop UAG Stop	UGA Stop UGG Trp	C A G
C	CUU CUC CUA CUG Leu	CCU CCC CCA CCG Pro	CAU His CAC CAA Gln CAG	CGU CGC CGA CGG Arg	U C A G
A	AUU AUC AUA Ile	ACU ACC ACA ACG Thr	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G
	AUG Met or start				
G	GUU GUC GUA GUG Val	GCU GCC GCA GCG Ala	GAU Asp GAC GAA Glu GAG	GGU GGC GGA GGG Gly	U C A G

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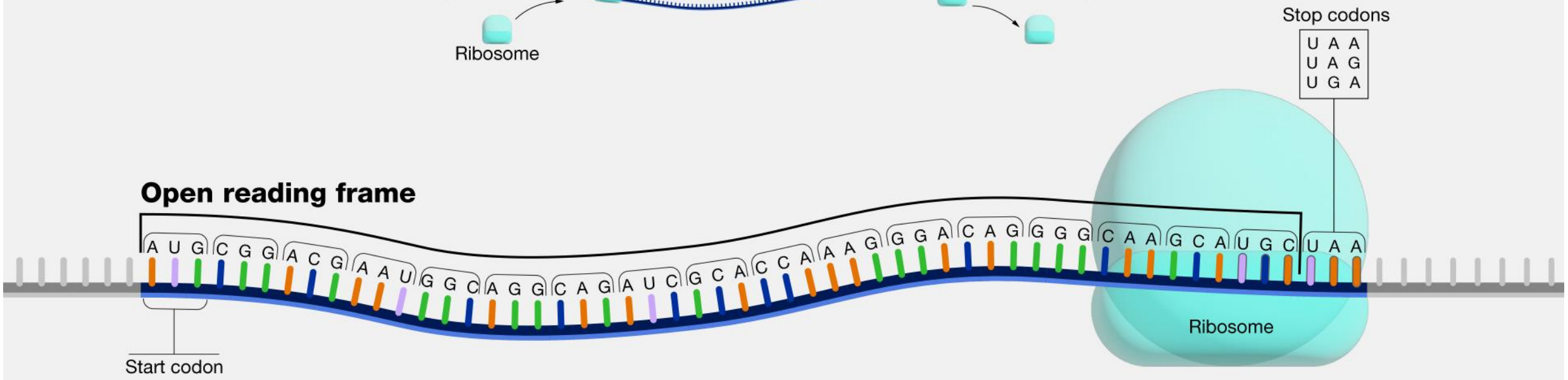
Open Reading Frame



First two letters are most important

Second mRNA base

First mRNA base (5′ end of codon)	U	C	A	G	Third mRNA base (3′ end of codon)
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met or start	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

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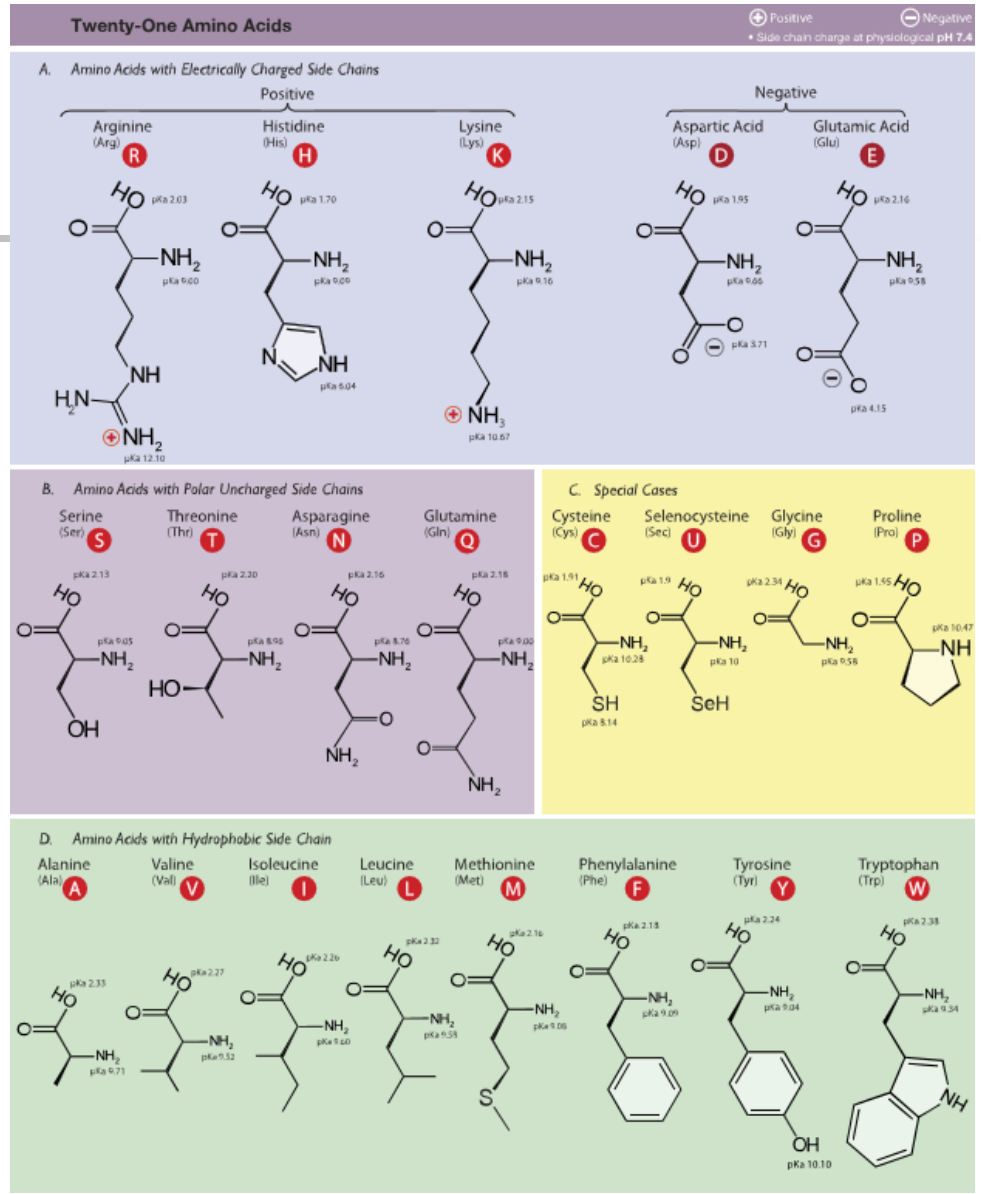
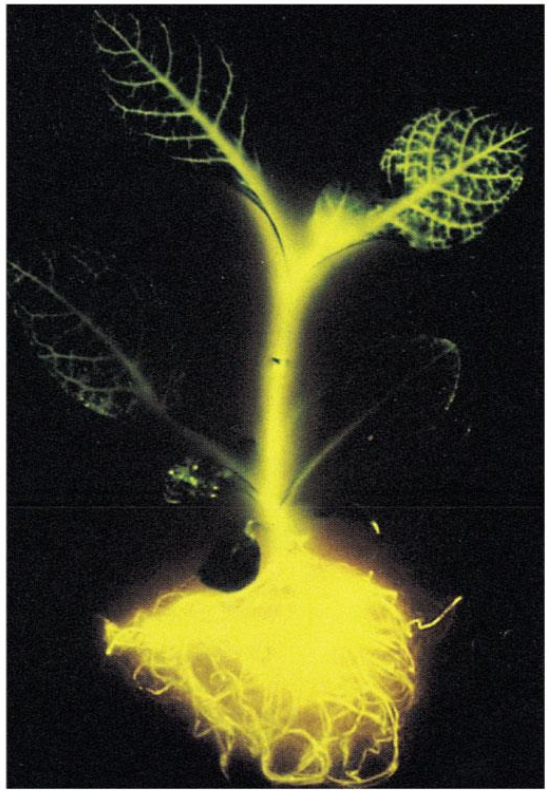


Genetic code universal



(a) Tobacco plant expressing a firefly gene

(b) Pig expressing a jellyfish gene

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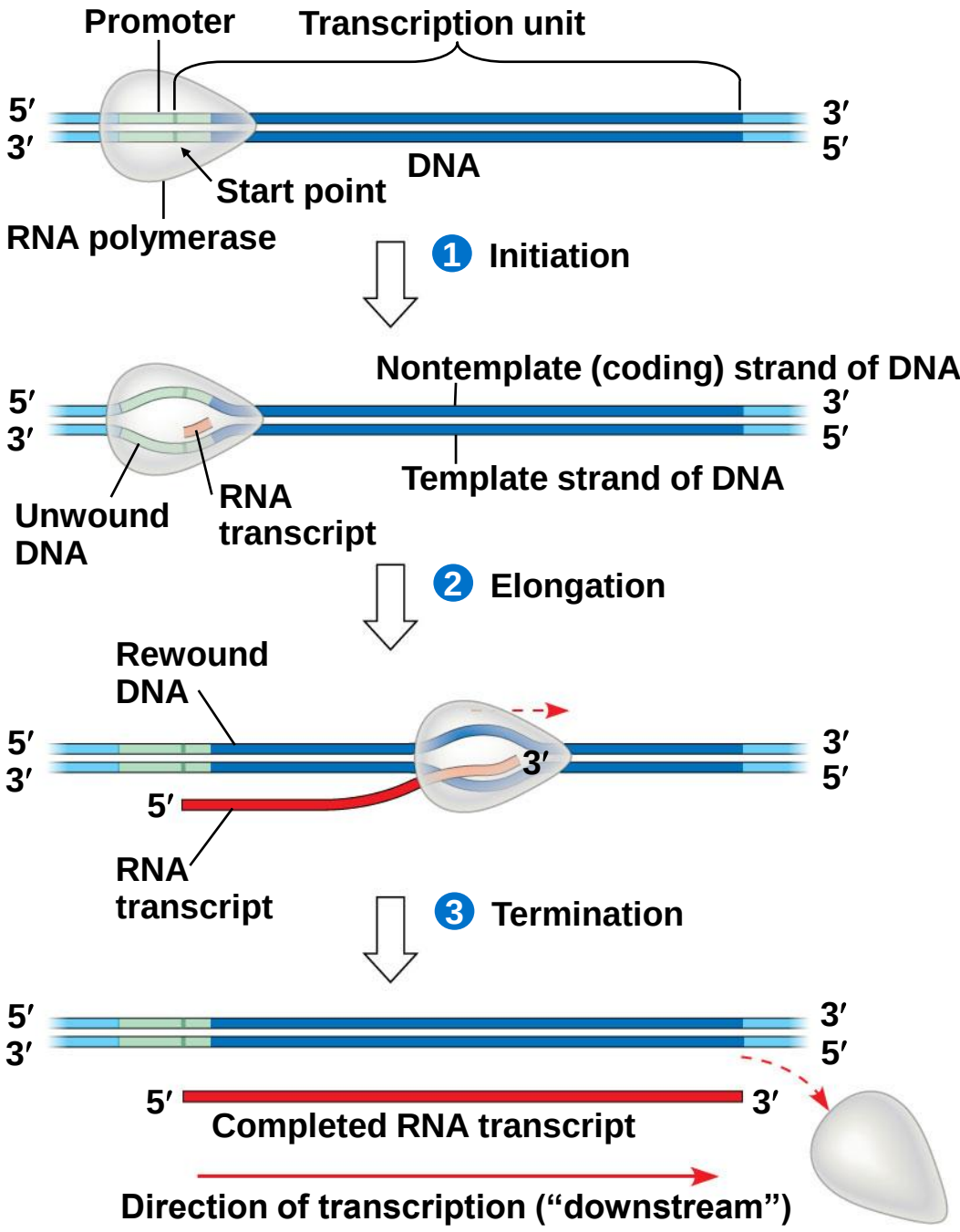
- **The genetic code is nearly universal**, shared by the simplest bacteria to the most complex animals (same ancestor?)
- **Genes** can be transcribed and translated after being **transplanted** from one species to another breaking the barriers between species (**Genetic engineering**)

基因在从一个物种移植到另一个物种后仍可被转录和翻译，从而打破了物种之间的**界限（基因工程）。**

Molecular Components of Transcription

- RNA synthesis is catalyzed by **RNA polymerase**, which pulls the DNA strands apart (10~20 bps) and hooks together the RNA nucleotides RNA聚合酶
- The RNA is complementary to the DNA template strand (AU, GC pairs)
- The DNA sequence where RNA polymerase attaches is called the promoter (100~1000 bps); in bacteria, the sequence signaling the end of transcription is called the terminator 启动子
- The stretch of DNA that is transcribed is called a transcription unit 转录单位



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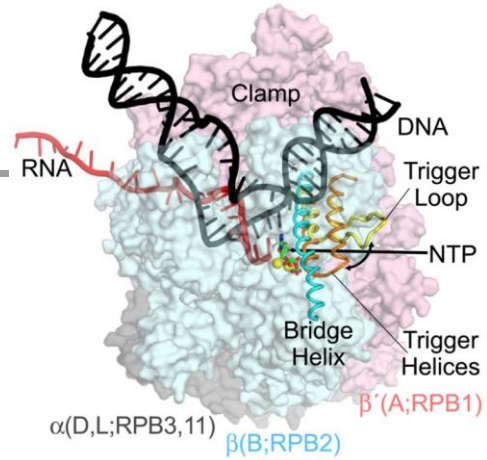
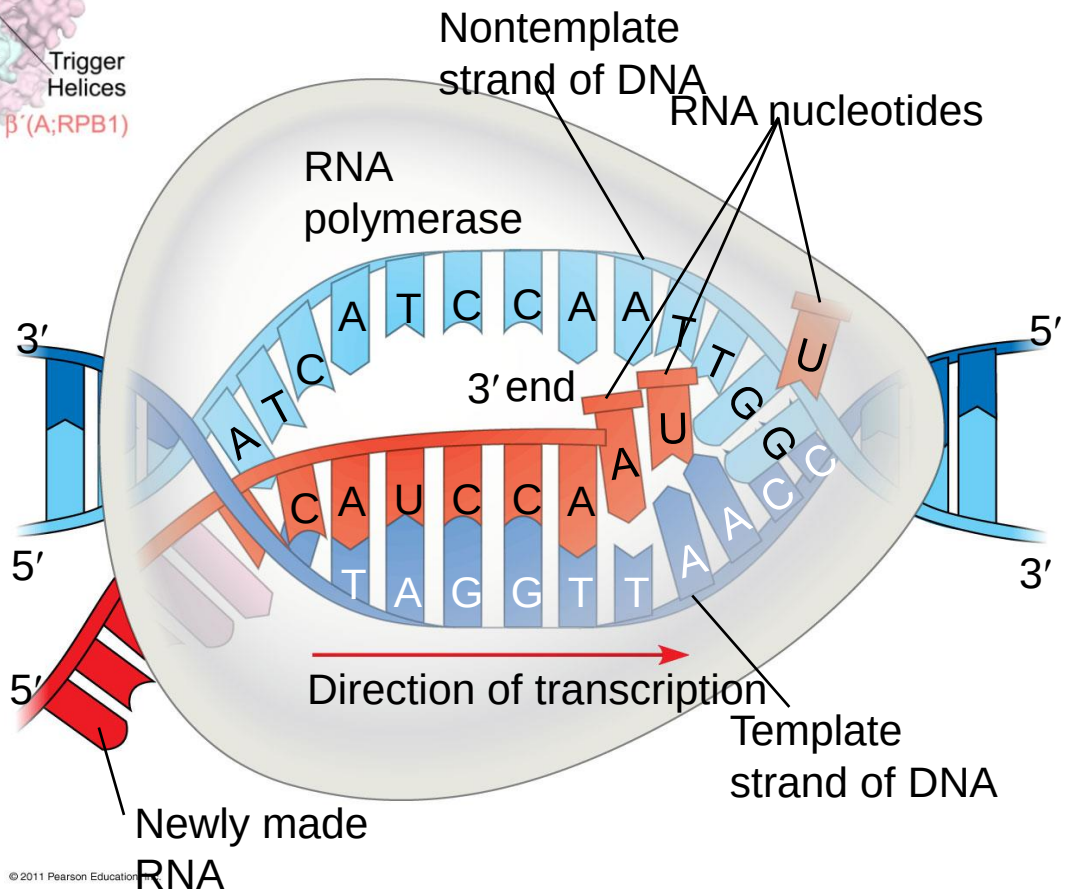
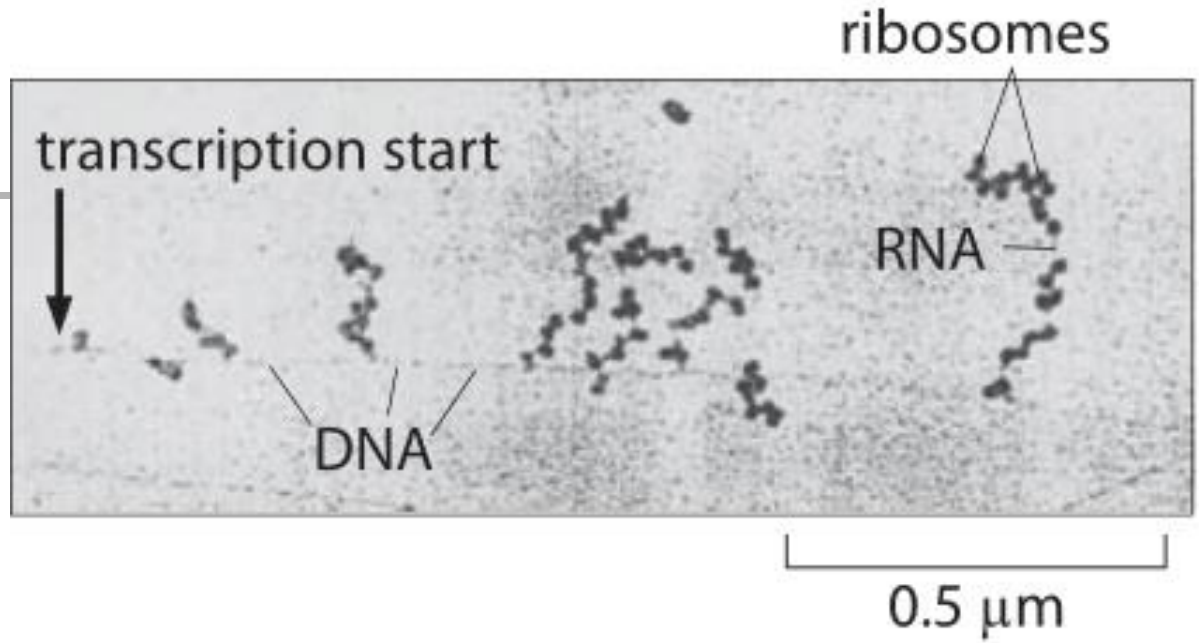
Transcription: Initiation, Elongation, Termination

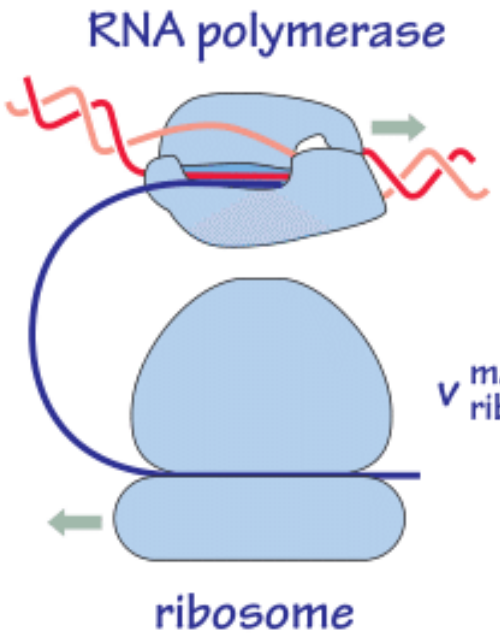
- As RNA polymerase moves along the DNA, it untwists the double helix, 10 to 20 bases at a time
- Transcription progresses at a rate of 40 nucleotides per second in eukaryotes
- A gene can be transcribed simultaneously by several RNA polymerases
- Nucleotides are added to the 3′ end of the growing RNA molecule







what is faster: transcription or translation?



$$v_{poly}^{max} \approx 40 - 80 \ \frac{nt}{s}$$

$$v_{ribo}^{max} \approx 20 \ \frac{aa}{s} = 60 \ \frac{nt}{s}$$

rates are similar!

Termination of Transcription

- The mechanisms of termination are different in bacteria and eukaryotes
- In bacteria, the polymerase stops transcription at the end of the terminator and the mRNA can be translated without further modification
- In eukaryotes, RNA polymerase II transcribes the polyadenylation (pol-As) signal sequence; the RNA transcript is released 10–35 nucleotides past this polyadenylation sequence

- Enzymes in the eukaryotic nucleus modify pre-mRNA (**RNA processing**) before the genetic messages are dispatched to the cytoplasm
- During RNA processing, both ends of the primary transcript are usually altered
- Also, usually some interior parts of the molecule are cut out, and the other parts spliced together

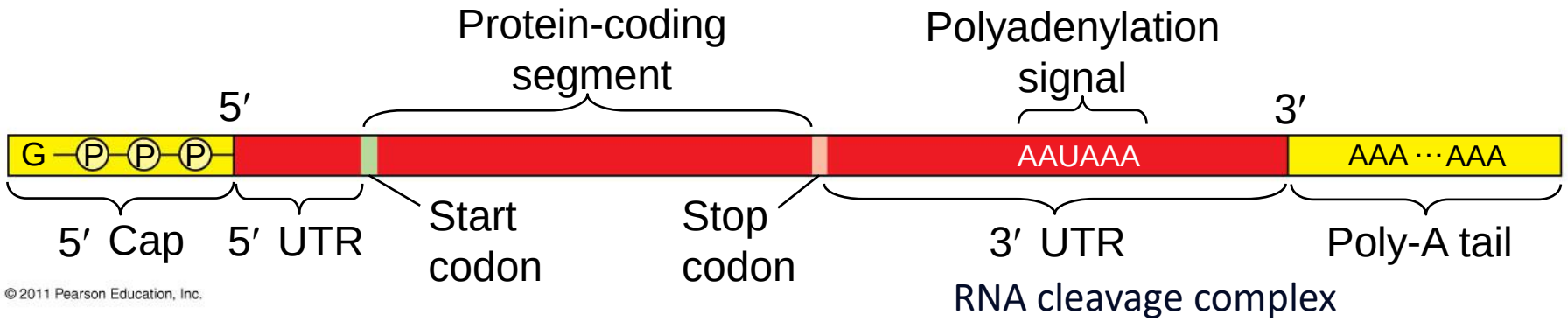
在真核生物中，RNA 聚合酶 II 转录多聚腺苷酸化（**多聚** A）**信号序列；**RNA 转录本在该多聚腺苷酸化序列后 10 **至** 35 个核苷酸处释放。真核细胞核内的酶在遗传信息被传递到细胞质之前对前体 mRNA（RNA 加工进行修饰。

Eukaryotic cells modify RNA after transcription

修饰

Alteration of mRNA Ends (Eukaryotic cells)

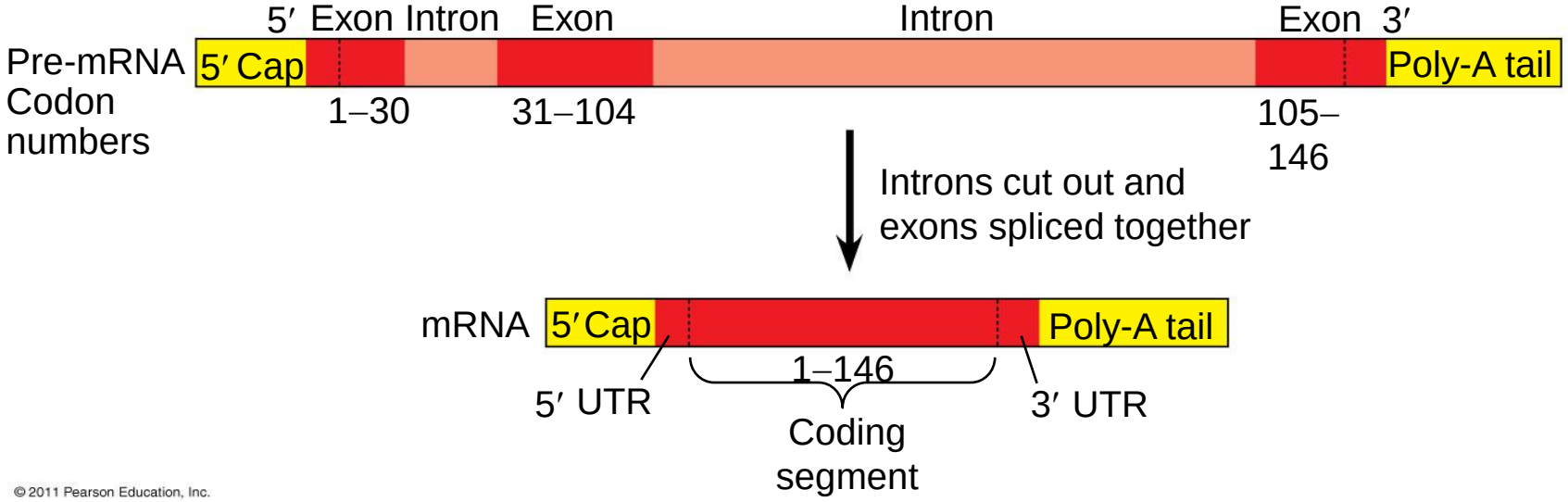
- Each end of a pre-mRNA molecule is modified in a particular way
  - The 5′ end receives a modified nucleotide **5′ cap**
  - The 3′ end gets a **poly-A tail**
- These modifications share several functions
  - They seem to facilitate the export of mRNA to the cytoplasm
  - They protect mRNA from hydrolytic enzymes
  - They help ribosomes attach to the 5′ end

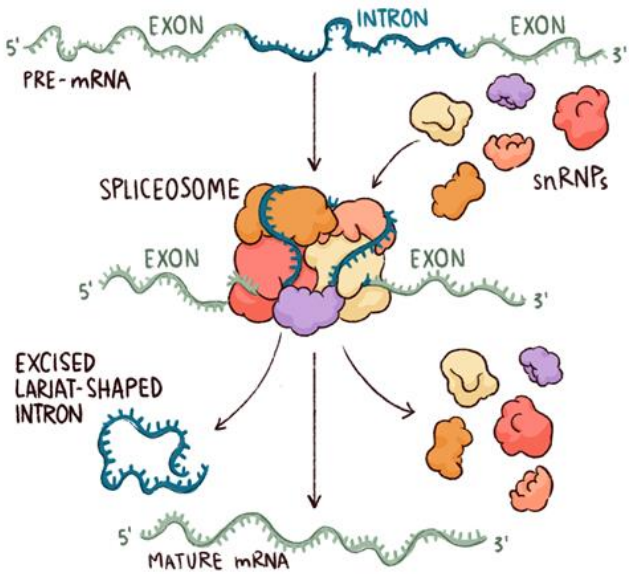


Untranslated region (UTR)

Split Genes and RNA Splicing

- Most eukaryotic genes and their RNA transcripts have long noncoding stretches of nucleotides that lie between coding regions
- These noncoding regions are called intervening sequences, or **introns** 内含子或间隔序列。
- The other regions are called **exons** because they are eventually expressed, usually translated into amino acid sequences 外显子
- **RNA splicing** removes introns and joins exons, creating an mRNA molecule with a continuous coding sequence RNA**剪接**
- In some cases, RNA splicing is carried out by spliceosomes(complex of sRNAs and 80 proteins) **剪接体**
- Spliceosomes consist of a variety of proteins and several small nuclear ribonucleoproteins (snRNPs) that recognize the splice sites (complex of snRNAs and 80 proteins)





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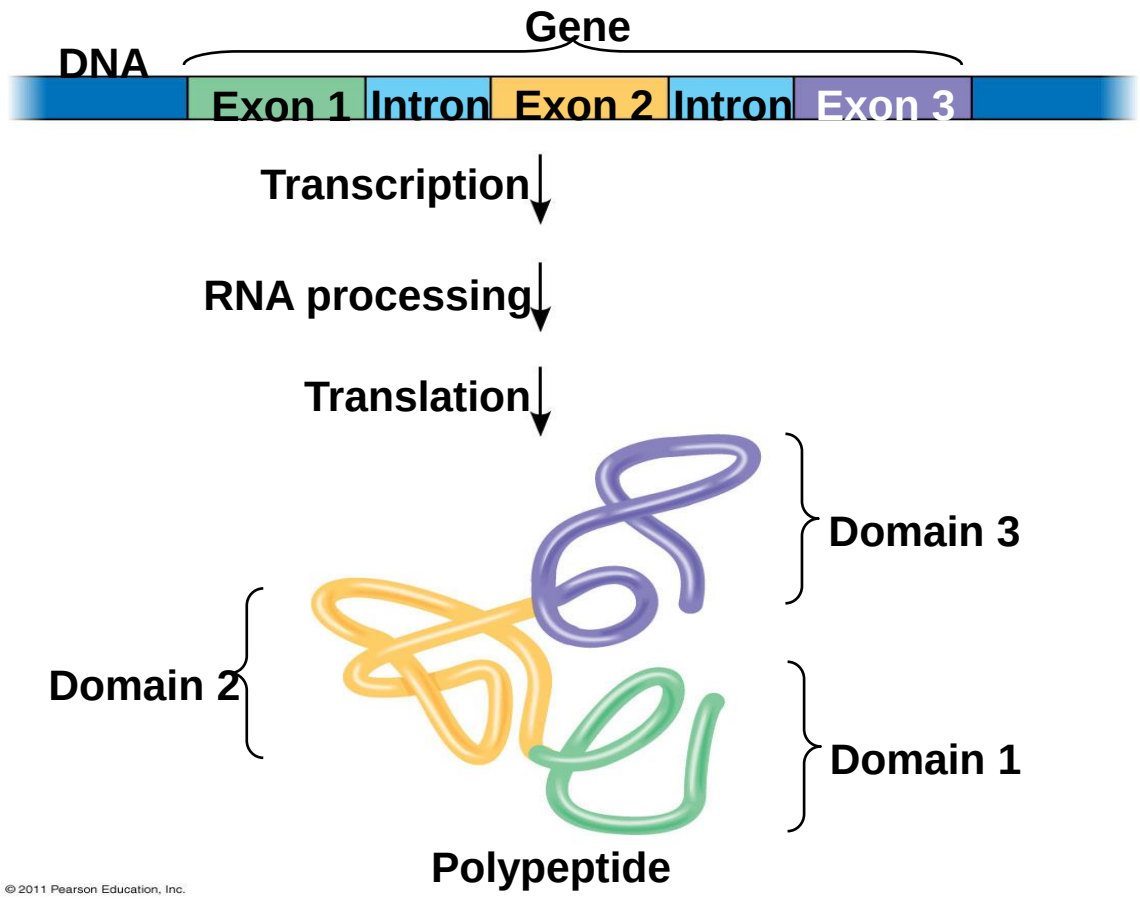
Protein domains

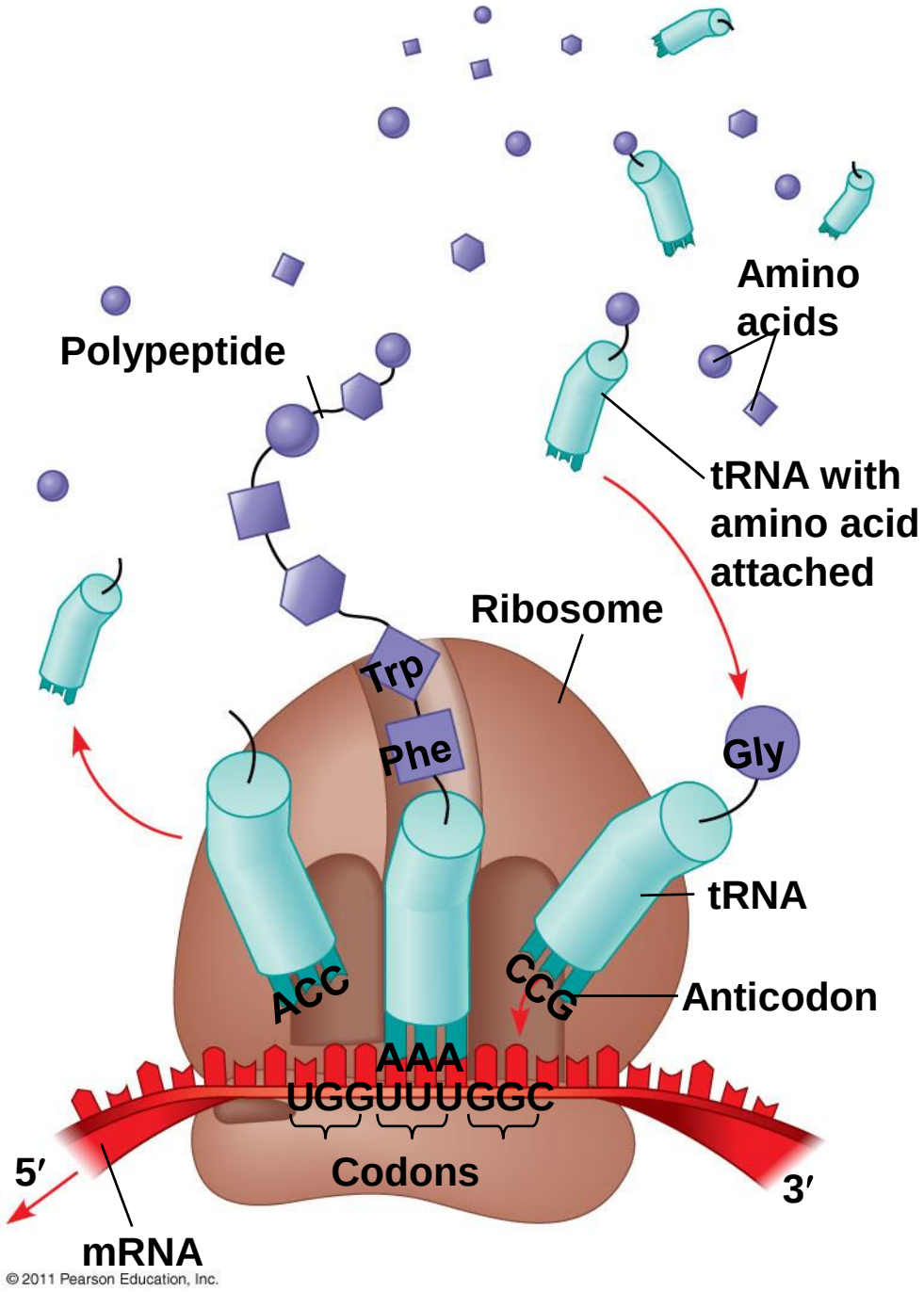
- Proteins often have a modular architecture consisting of discrete regions called **domains**
- In many cases, different exons code for the different domains in a protein
- Exon shuffling may result in the evolution of new proteins 外显子重排



Domains often form functional units (structure-functional modules), and they are independently stable. Alpha-fold.3 may design proteins that never seen before (1billion years evolution, Nature, 2024))

Molecular Components of Translation

- A cell translates an mRNA message into protein with the help of **transfer RNA (tRNA)** 转运核糖核酸
- tRNAs transfer amino acids to the growing polypeptide in a ribosome
- Translation is a complex process in terms of its biochemistry and mechanics
- tRNAs are transcribed by RNA polymerase III as pre-tRNAs in the nucleus. (about 500 tRNA genes in human cells)



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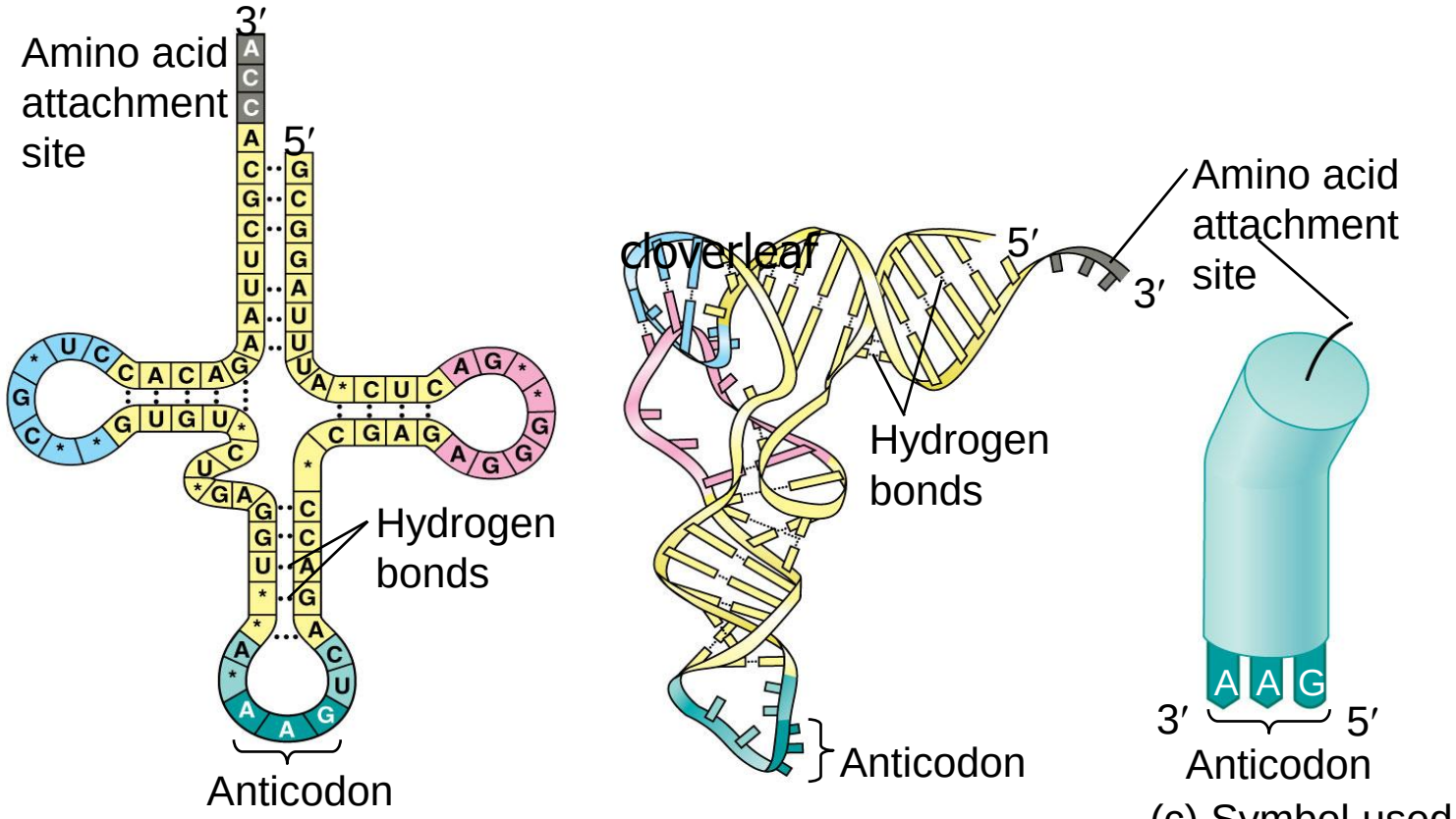
The structure of transfer RNA (tRNA).

不完全相同



(a) Two-dimensional structure (b) Three-dimensional structure (c) Symbol used in this book

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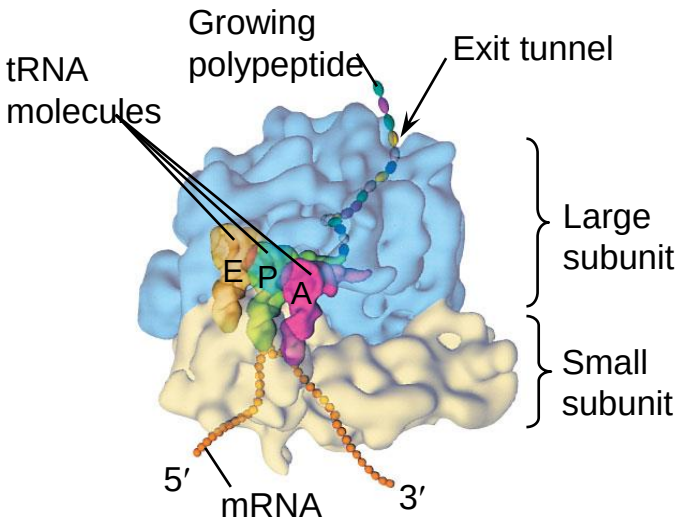
- Molecules of tRNA are not identical
- Each carries a specific amino acid on one end (done by aminoacyl-tRNA synthetase)
- Each has an **anticodon** on the other end; the anticodon base-pairs with a complementary codon on mRNA
- A tRNA molecule consists of a single RNA strand that is only about 80 nucleotides long
- Flattened into one plane to reveal its base pairing, a tRNA molecule looks like a cloverleaf

核糖体的两个亚基（大亚基和小亚基）
由蛋白质和核糖体 RNA（rRNA）**构成。**

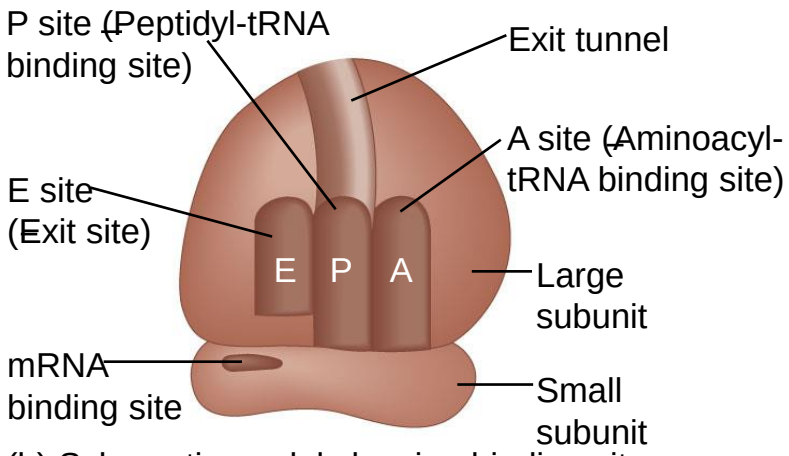
Ribosomes

- Ribosomes facilitate specific coupling of tRNA anticodons with mRNA codons in protein synthesis
- The two ribosomal subunits (large and small) are made of proteins and **ribosomal RNA (rRNA)**
- Bacterial and eukaryotic ribosomes are somewhat similar but have significant differences: some antibiotic drugs specifically target bacterial ribosomes without harming eukaryotic ribosomes

核糖体在蛋白质合成过程中促进 tRNA 反密码子与 mRNA 密码子的特异性配对



(a) Computer model of functioning ribosome



(b) Schematic model showing binding sites

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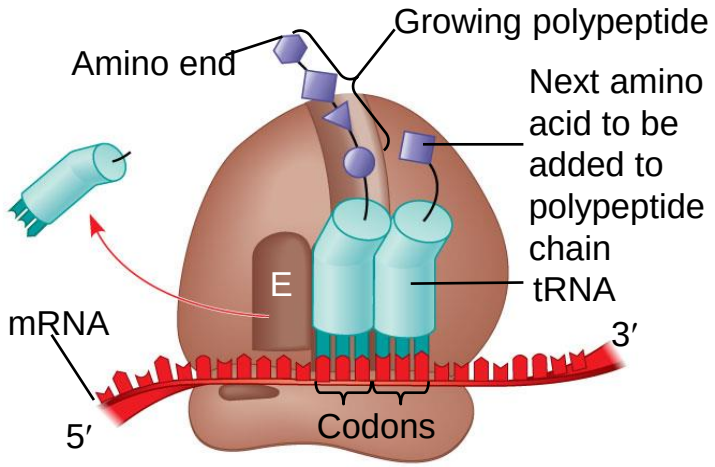
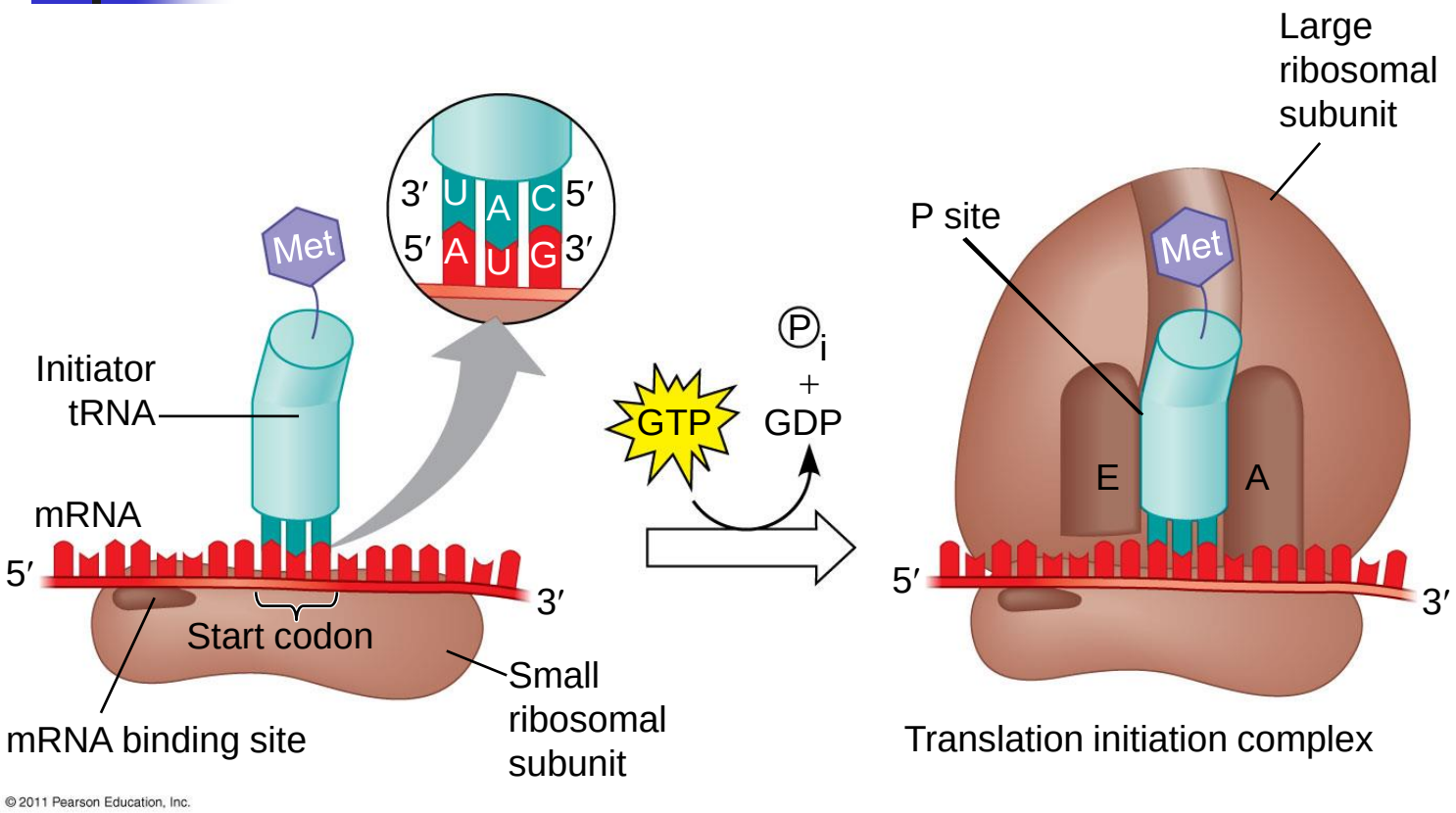


(c) Schematic model with mRNA and tRNA

Elongation of the Polypeptide Chain



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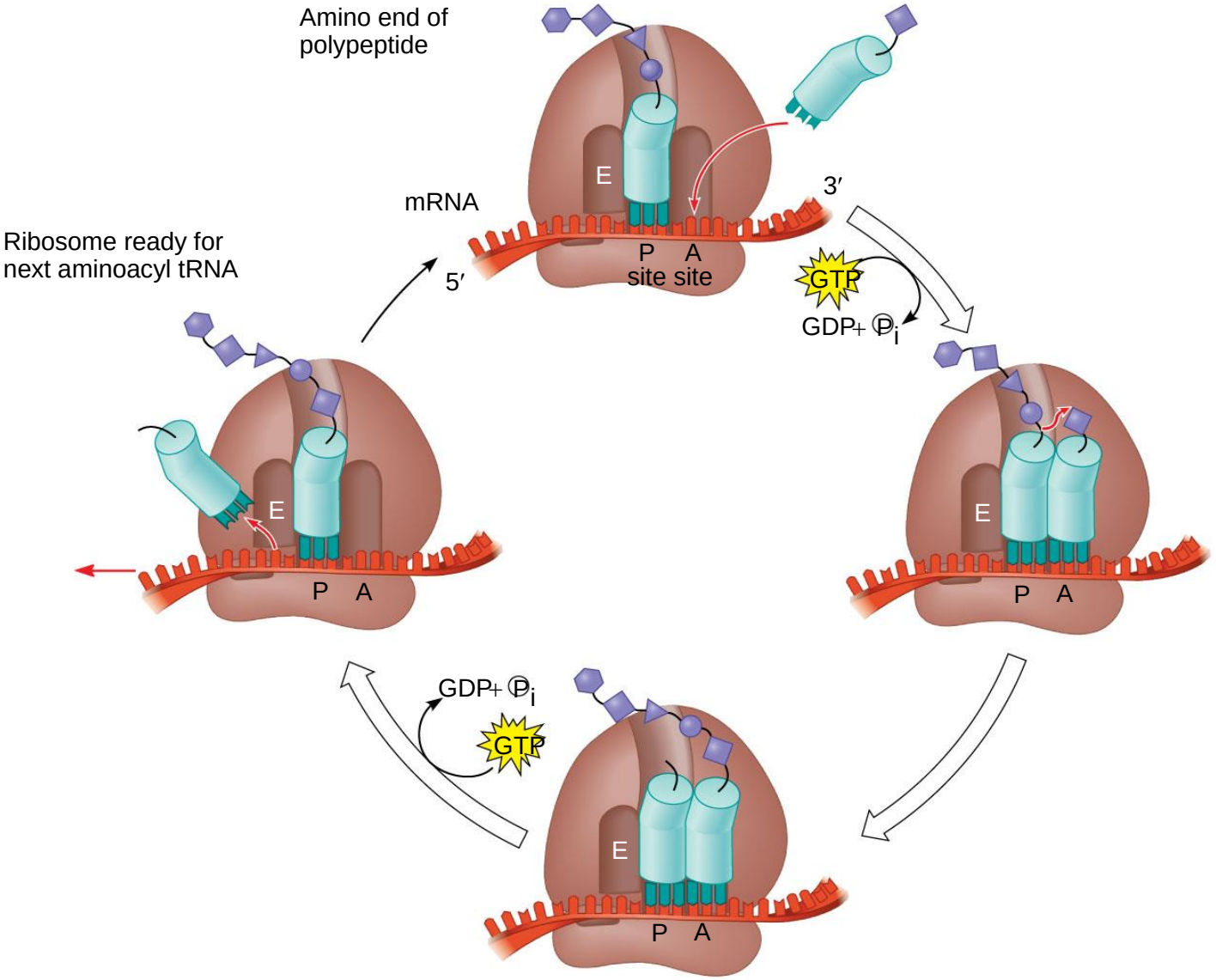
延伸阶段

- During the elongation stage, amino acids are added one by one to the preceding amino acid at the **C-terminus** of the growing chain
- Each addition involves proteins called elongation factors and occurs in three steps: codon recognition, peptide bond formation, and translocation
- Translation proceeds along the mRNA in a 5′ to 3′ direction

nucleotide guanosine triphosphate (GTP)



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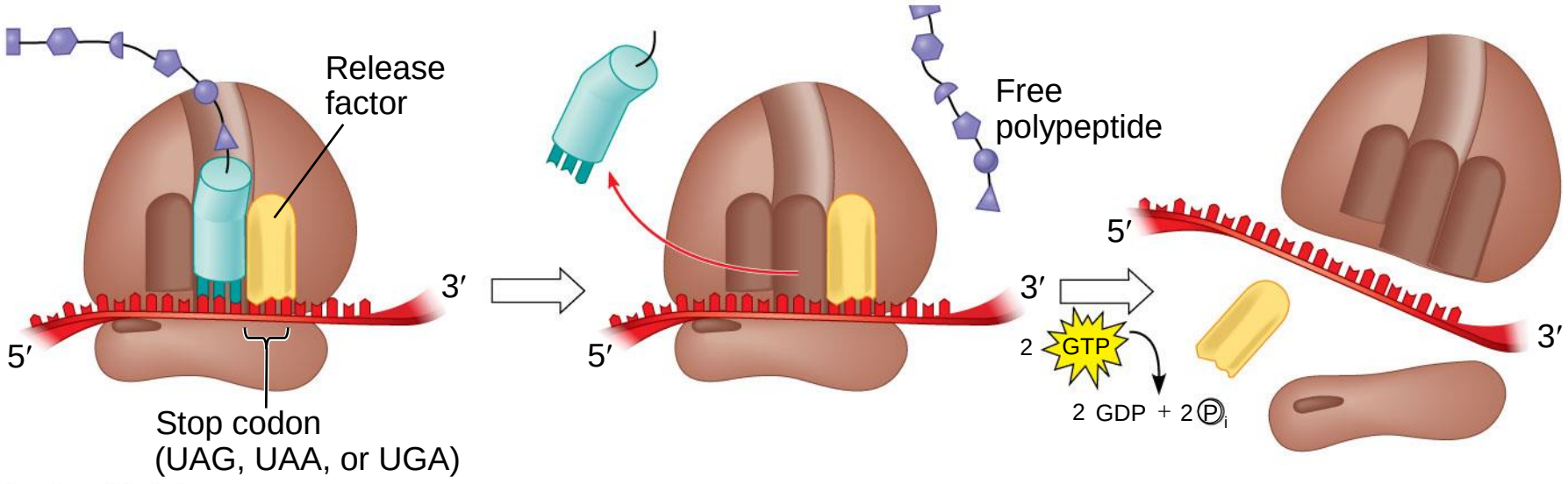
Termination of Translation

- Termination occurs when a stop codon in the mRNA reaches the A site of the ribosome
- The A site accepts a protein called a release factor (a protein) 释放因子
- The release factor causes the addition of a water molecule instead of an amino acid
- This reaction releases the polypeptide, and the translation assembly then comes apart



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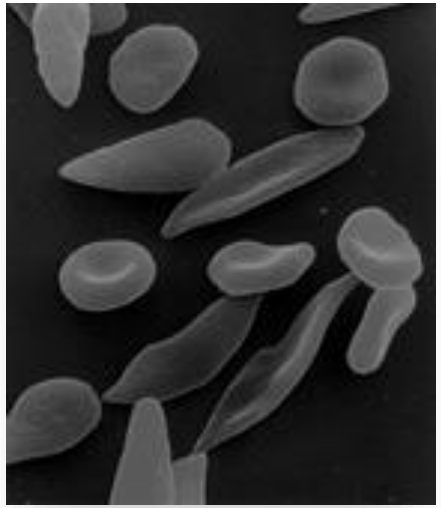
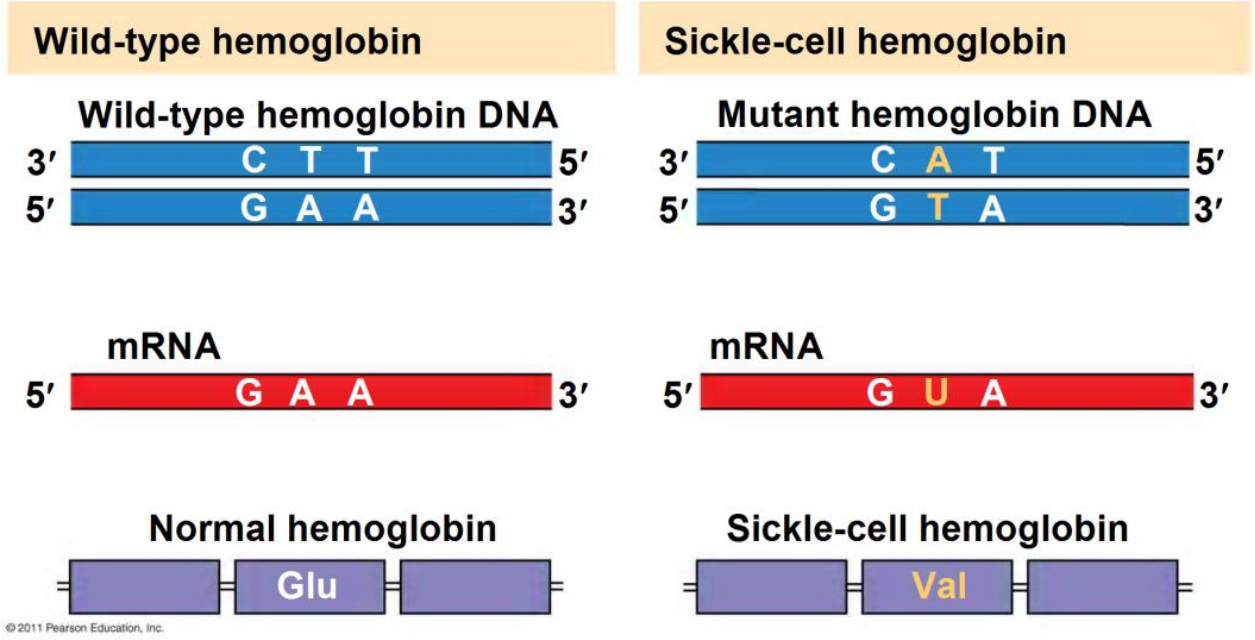
Mutations of one or a few nucleotides can affect protein structure and function



- **Mutations** are changes in the genetic material of a cell or virus
- **Point mutations** are chemical changes in just one base pair of a gene
- The change of a single nucleotide in a DNA template strand can lead to the production of an abnormal protein

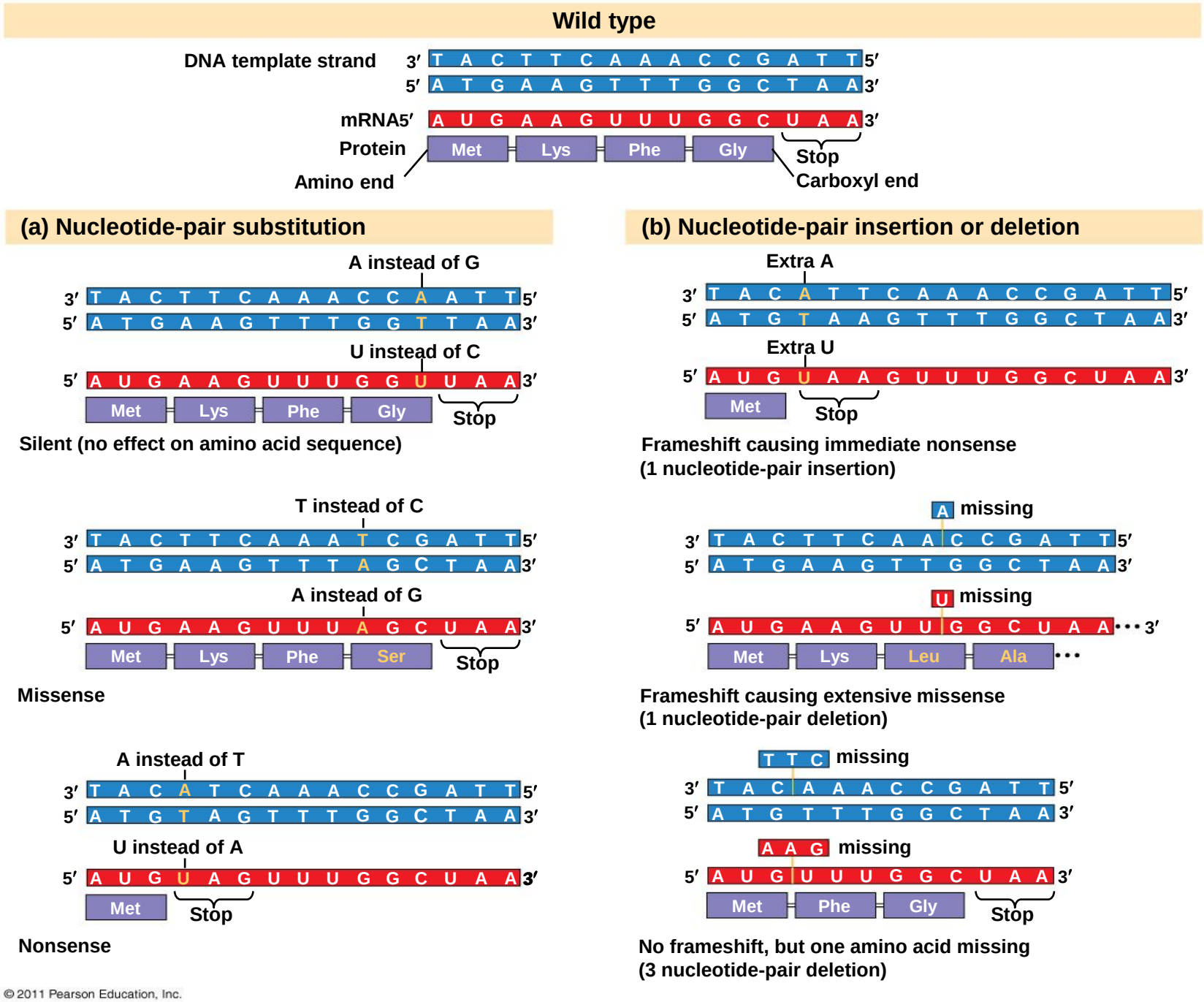
The molecular basis of sickle-cell disease: a point mutation.

- Sickled cells are more rigid and can get stuck and block blood vessels, which stops the oxygen from getting through. That can cause a lot of pain. It can also harm organs, muscles, and bones.



Types of Small-Scale Mutations

- Point mutations within a gene can be divided into two general categories
  - Nucleotide-pair *Substitutions and insertion/deletion*
- A **nucleotide-pair substitution** replaces one nucleotide and its partner with another pair of nucleotides
- **Silent mutations** have no effect on the amino acid produced by a codon because of redundancy in the genetic code
- **Missense mutations** still code for an amino acid, but not the correct amino acid
- **Nonsense mutations** change an amino acid codon into a stop codon, nearly always leading to a nonfunctional protein



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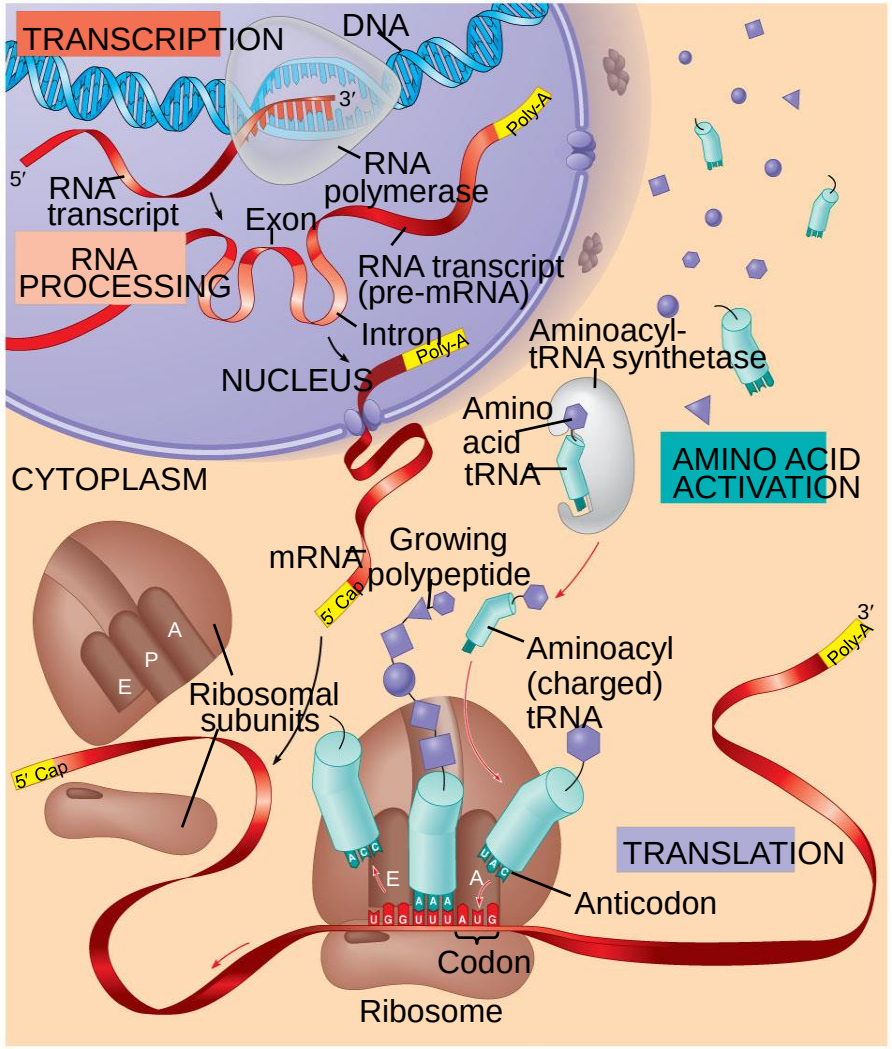
Insertions and Deletions

- **Insertions** and **deletions** are additions or losses of nucleotide pairs in a gene
- These mutations have a disastrous effect on the resulting protein more often than substitutions do
- Insertion or deletion of nucleotides may alter the reading frame, producing a **frameshift mutation** 移码突变
- Spontaneous mutations can occur during DNA replication, recombination, or repair
- Mutagens are physical or chemical agents that can cause mutations
- The idea of the gene has evolved through the history of genetics
- We have considered a gene as
  - A discrete unit of inheritance
  - A region of specific nucleotide sequence in a chromosome
  - A DNA sequence that codes for a specific polypeptide chain
  - ("RNA" coding now??)

Revision



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DNA Learning Center

CSH Cold Spring Harbor Laboratory

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Polymerase chain reaction (PCR)

DNA引物

- Although **primers** composed of **RNA** can be used, **DNA primers** are more commonly seen in **PCR** as they are more temperature-stable, which is important during the annealing process

PCR.mp4

Satellite DNA consists of very large arrays of tandemly repeating (**STR**), non-coding DNA. Such as CT:AG, CCT:AGG, CTT:AAG, and

TCCC TCCC TCCC TCCC TCCCT
AGGGAGGGAGGGAGGGAGGG

DNA forensic test

- Design Primers
- Run PCR
- Run Electrophoresis

Short Tandem Repeats (STR)

THO1 6, 8

➡ 6 ➡ AATG repeated 6 times

➡ 8 ➡ AATG repeated 8 times

One repeat from each parent

TH01 is a tetrameric short tandem repeat locus located in intron 01 of the tyrosine hydroxylase gene.

Short Tandem Repeats (STR)

U.S. ➡ 13

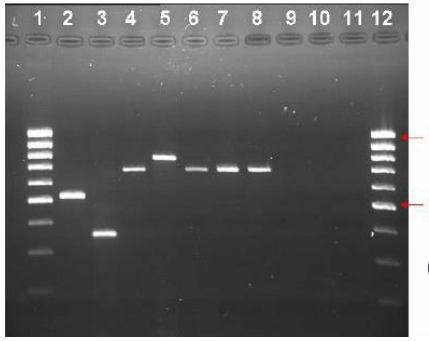
U.K. ➡ 10

STRs	Probability
3	1 in 5000
6	1 in 2 000 000
9	1 in 1 000 000 000
13	1 in 100s of trillions

Child STRs inherited half from mother and another half from father. That is why some of bands is missing from child's STRs.

An electrophoresis apparatus running five sample of DNA

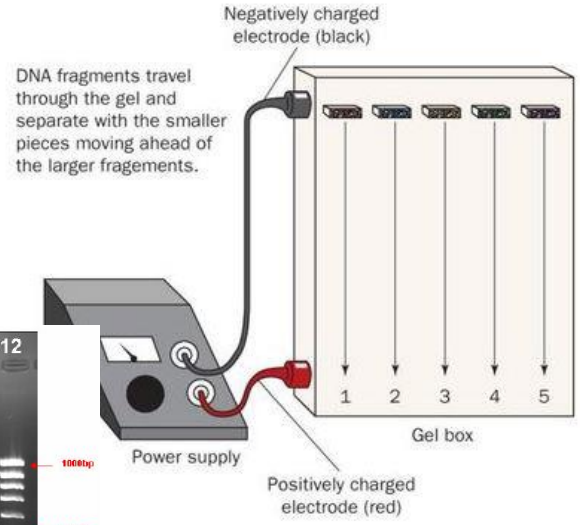
Arrows show the movement of the negatively charged DNA fragments through the gel matrix



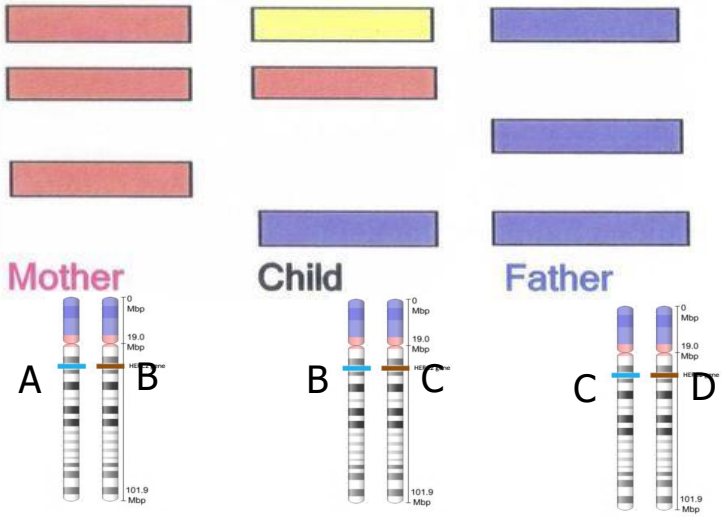
nitrogenous base

UV light

ethidium bromide,



Bands and widths are significant in matching samples of DNA



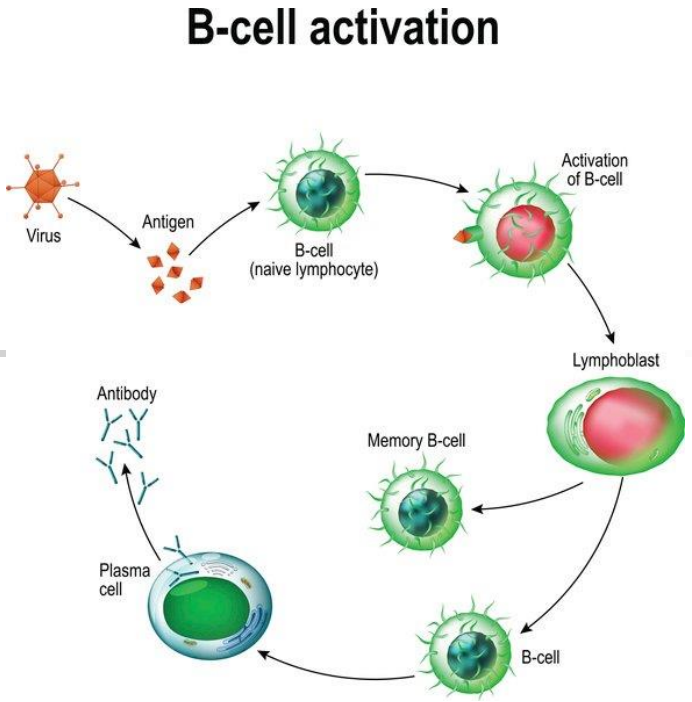
RNA vaccine

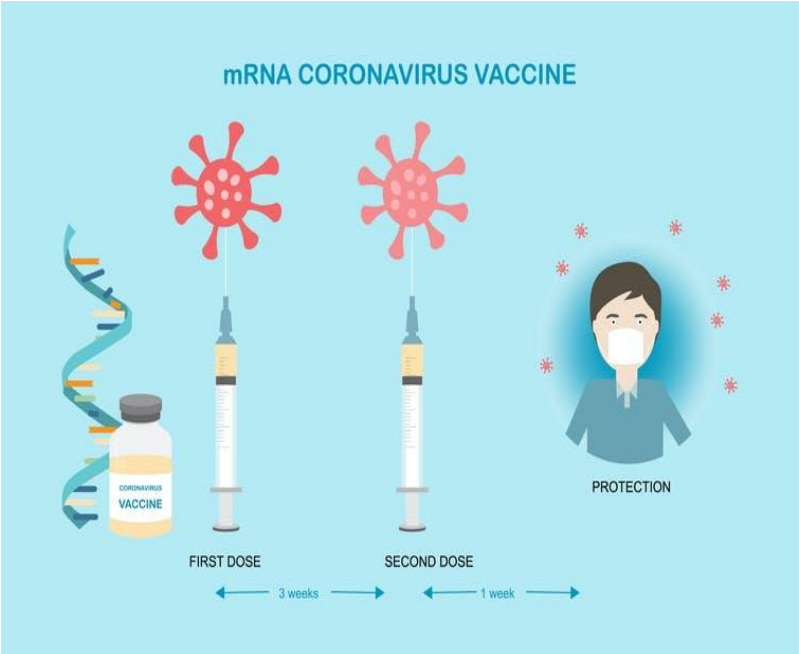
- **How does mRNA vaccine work?**
  - Vaccines train the immune system to recognize the disease-causing part of a virus. Vaccines traditionally contain either weakened viruses or purified signature proteins of the virus.
  - But an mRNA vaccine is different, because rather than having the viral protein injected, a person receives genetic material – mRNA
  - While this synthetic mRNA is genetic material, it cannot be transmitted to the next generation. After an mRNA injection, this molecule guides the protein production inside the muscle cells
  - **What are problems with mRNA?**

The most important challenge for development of a mRNA vaccine remains its inherent instability, because it is more likely to break apart above freezing temperatures. But beginning in 2005, researchers figured out how to stabilize mRNA and package it into small particles to deliver it as a vaccine

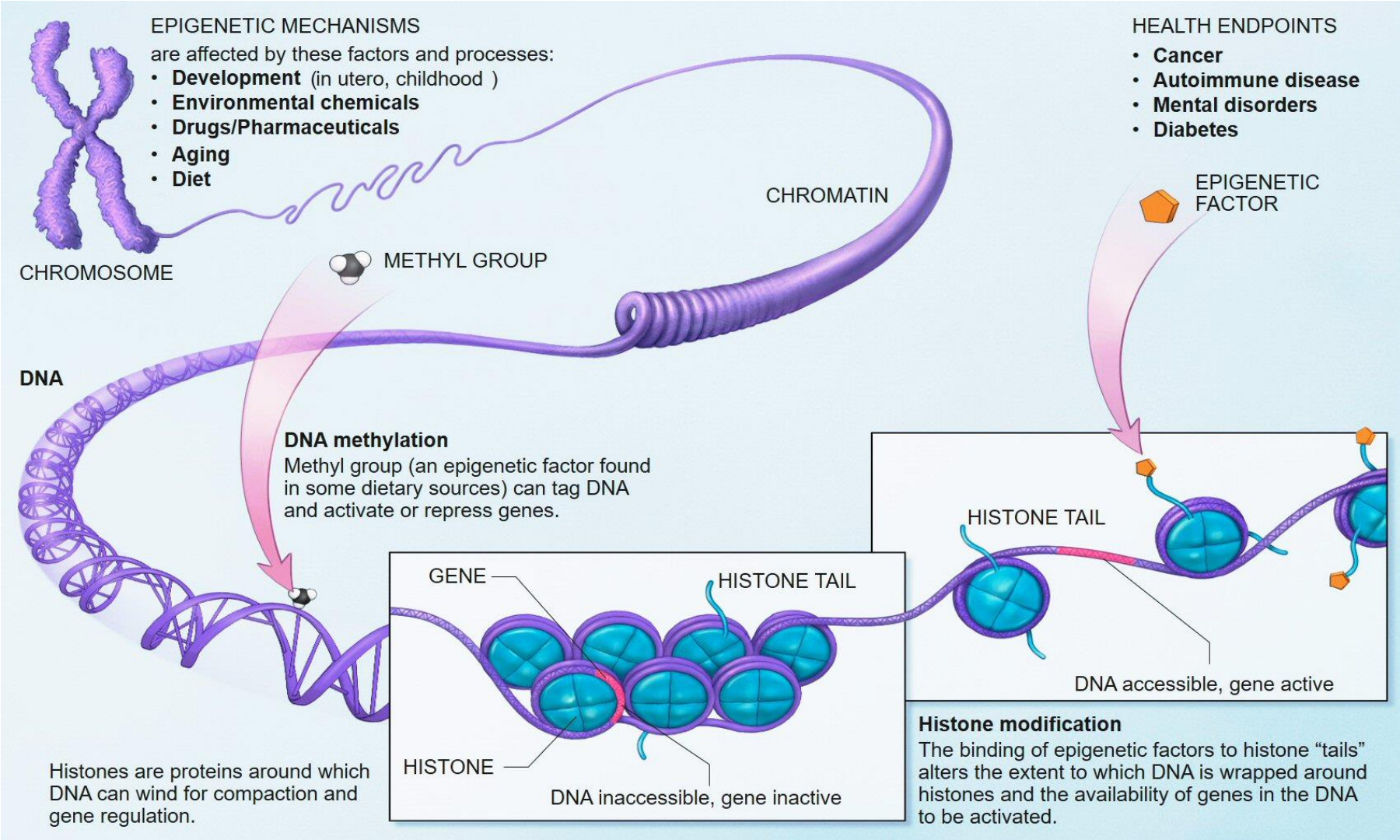
抗原

"**Antigens**" are molecular structures on the surface of viruses that are recognized by the immune system and are capable of triggering one kind of immune response known as antibody production.



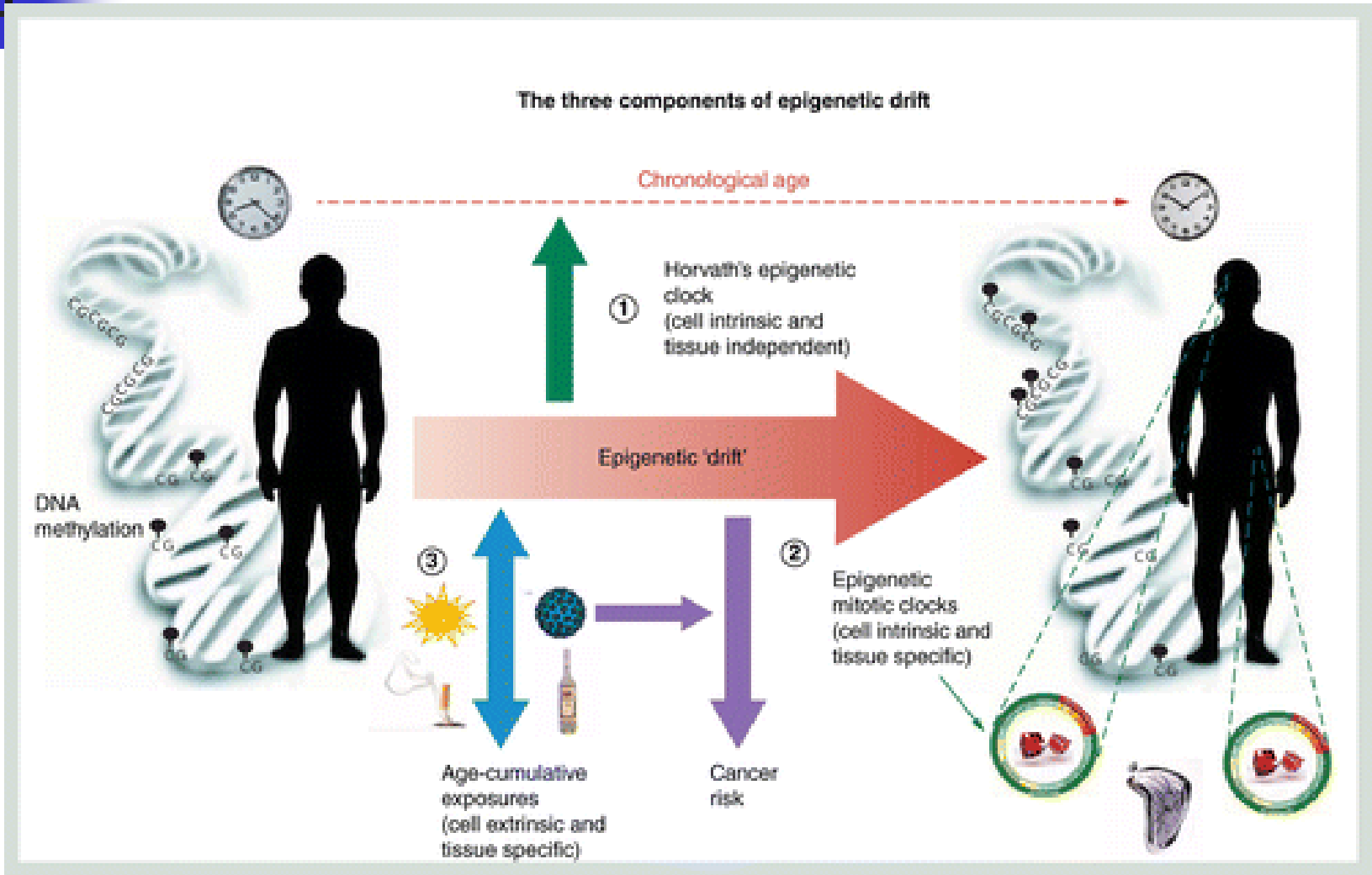


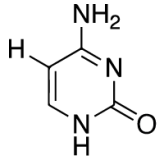
Epigenetics 表观遗传学



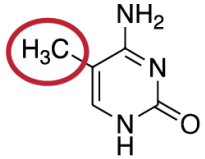
Epigenetics most often involves changes that affect gene activity and expression, but the term can also be used to describe any heritable phenotypic change. Such effect on cellular and physiological phenotypic traits may result from external or environmental factors, or be part of normal development.

Epigenetic clock based on DNA methylation levels

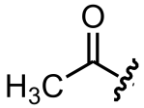




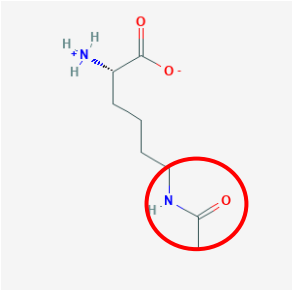
Cytosine



methylated Cytosine



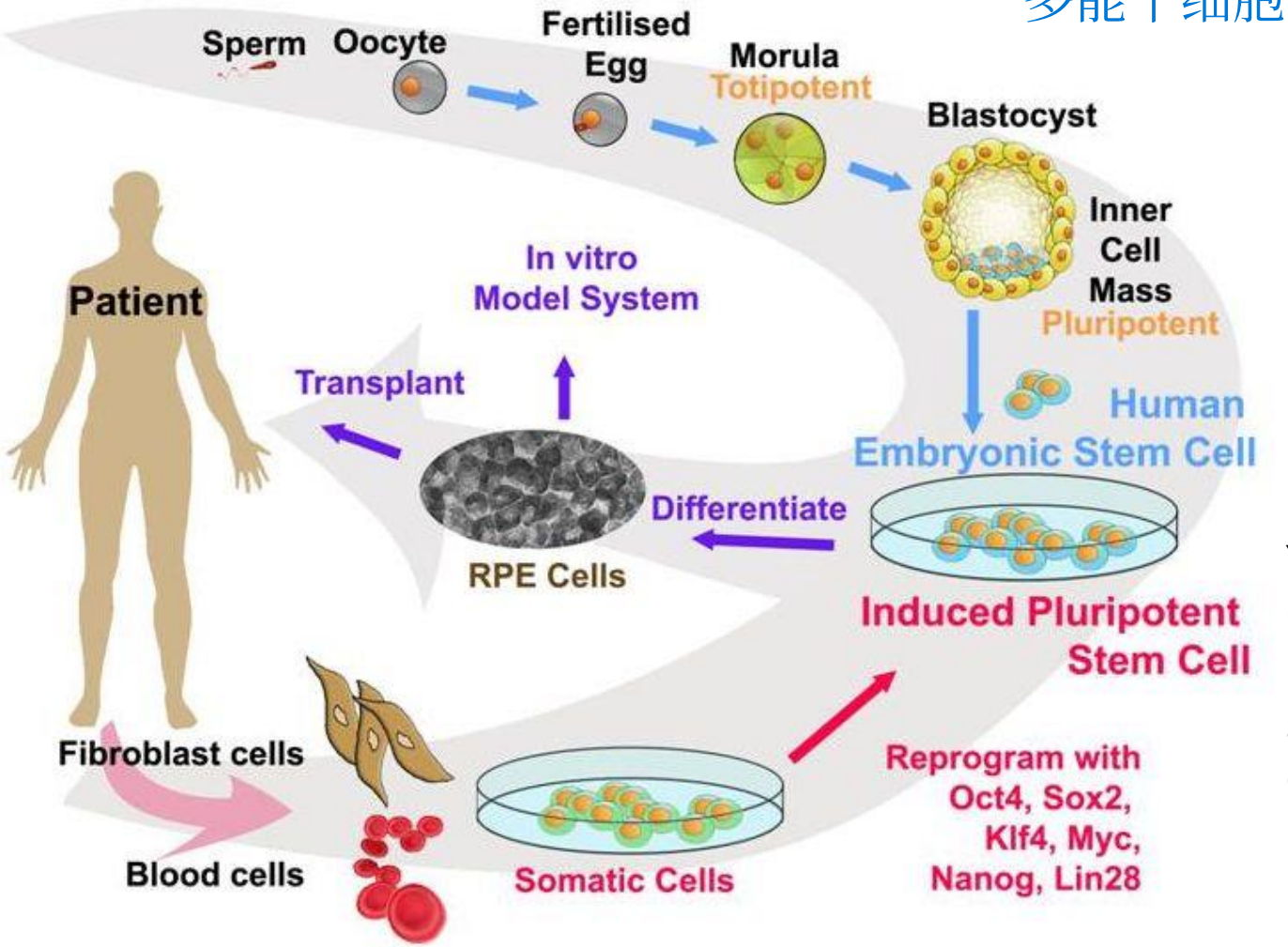
acetyl group



Acetyl-lysine

Induced Pluripotent Stem Cells (iPSC): Meaning, Function and Significance

多能干细胞



成纤维细胞是一种有助于结缔组织形成的细胞。体细胞是除精子和卵细胞之外身体内的细胞。能够发育成体内多种不同细胞或组织的干细胞。

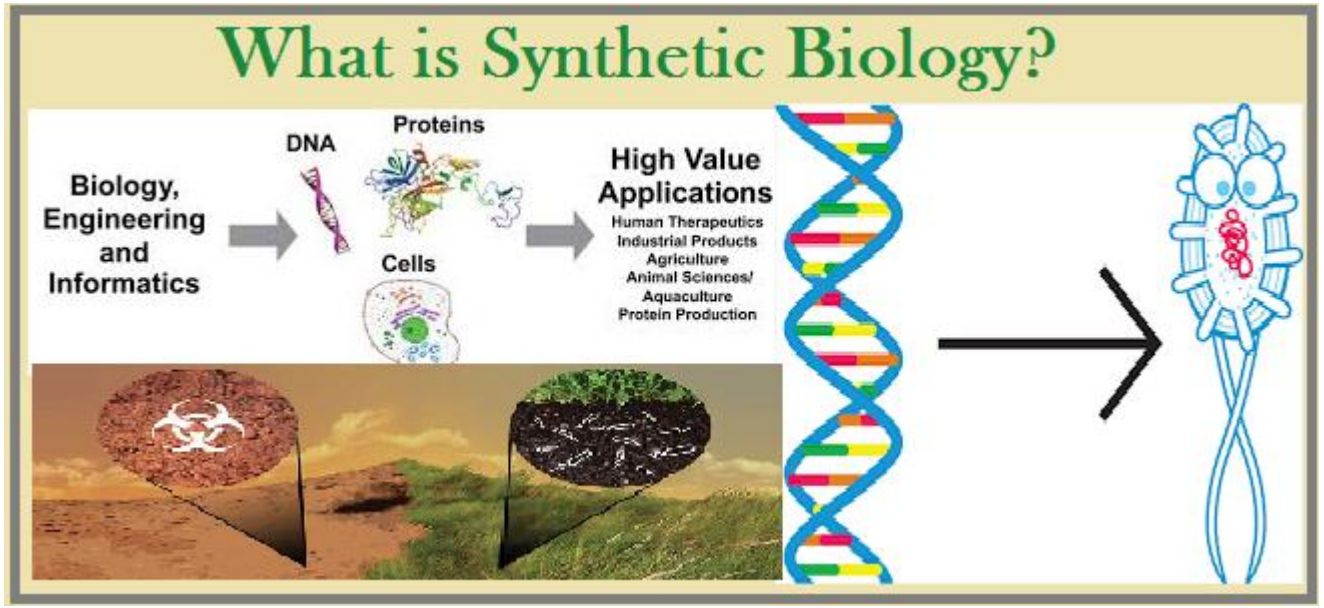
Yamanaka S

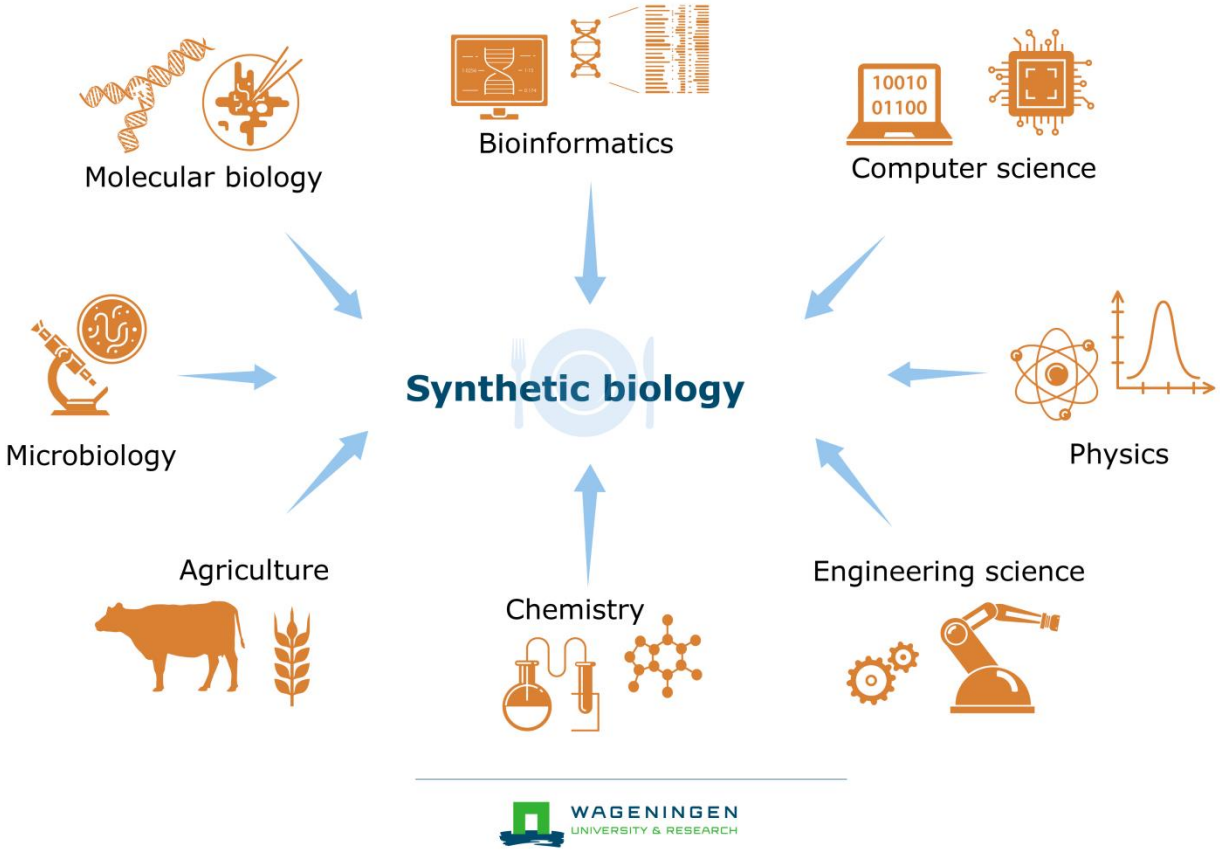
Oct3/4, Sox2, Klf4, and c-Myc. Human iPS cells were similar to human embryonic stem (ES) cells in morphology, proliferation

A fibroblast is a type of cell that contributes to the formation of connective tissue

Somatic cells are the cells in the body other than sperm and egg cells, A stem cell that is able to develop into many different types of cells or tissues in the body

Synthetic Biology



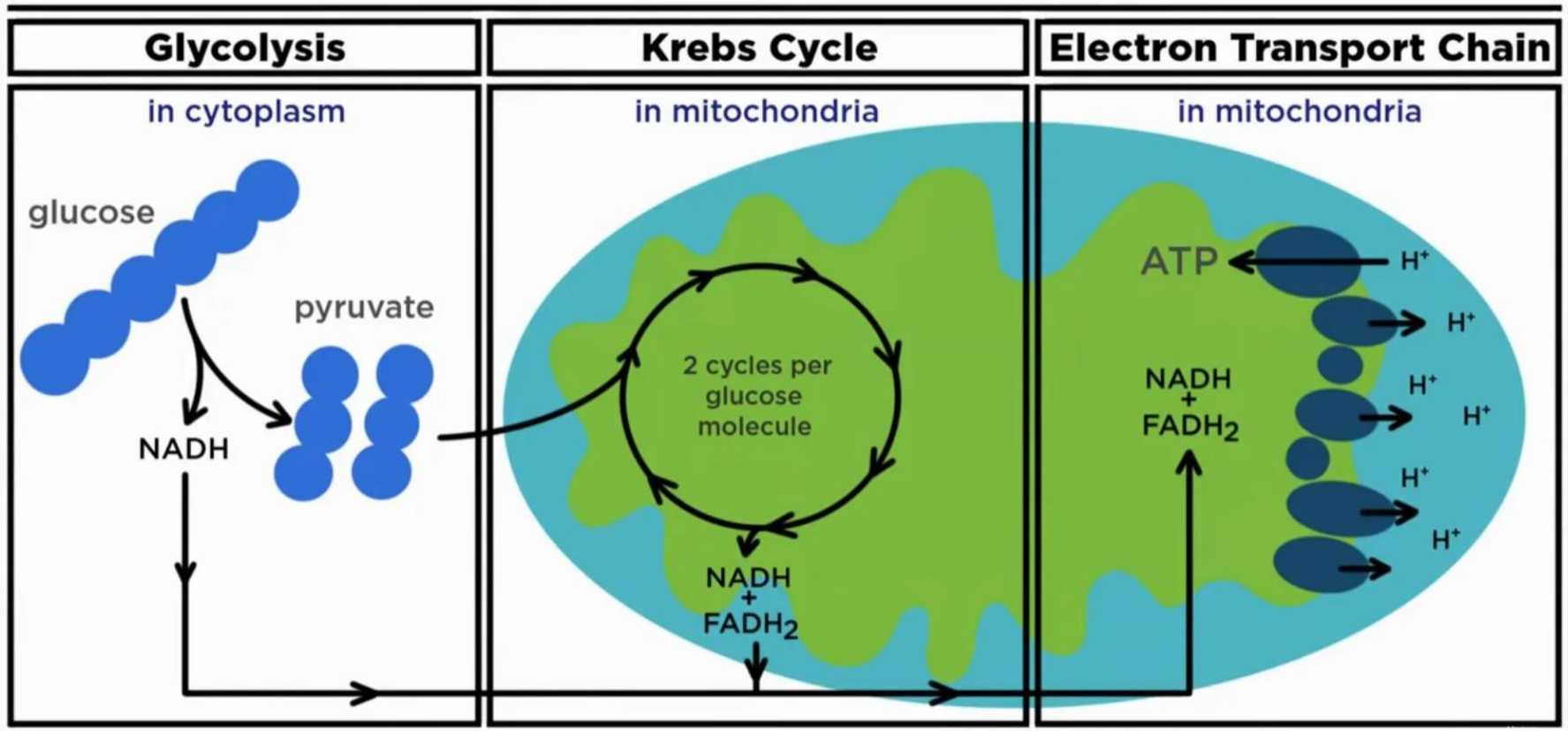


重组蛋白生产

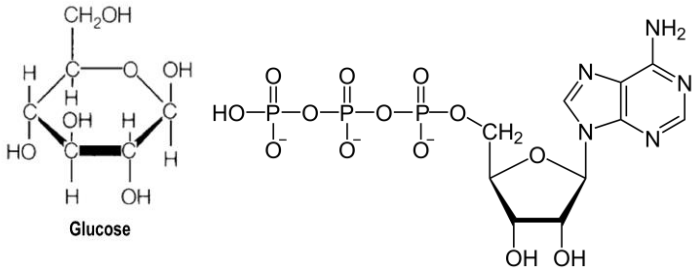
Recombinant protein production

琼脂

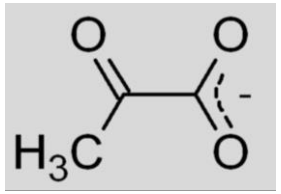
- **agar** is a mixture of two components, the linear polysaccharide agarose and a heterogeneous mixture of smaller molecules called agaropectin



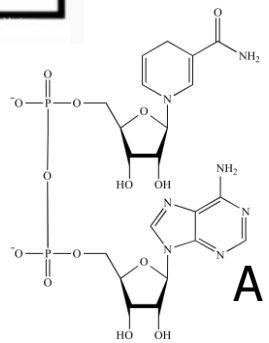
These processes requires B3 and B2. NADH (Nicotinamide adenine dinucleotide) is a derivative of Vitamin B3 (Niacin/Nicotinamide) while FADH2 is a derivative of Vitamin B2 (Riboflavin). This is the key difference between NADH and FADH2.



ATP

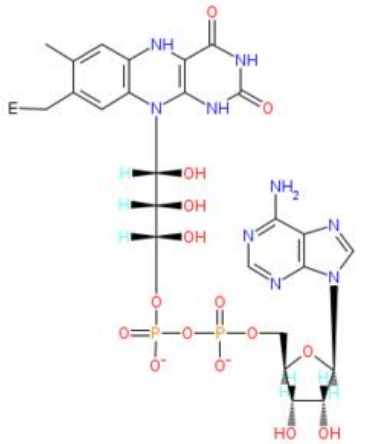


pyruvate



A

NADH

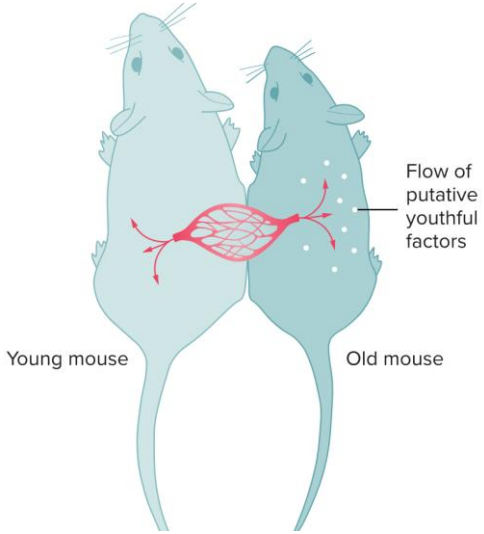


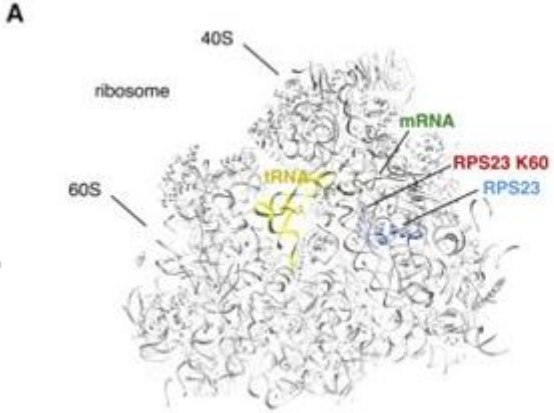
A

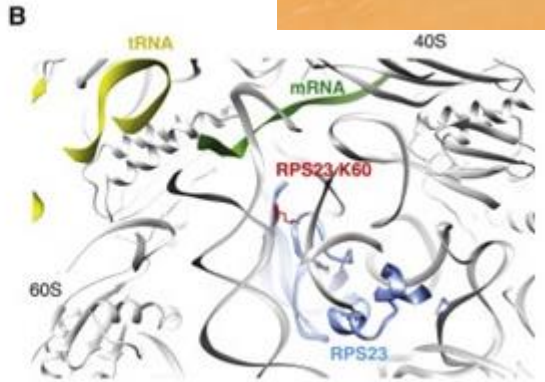
FADH

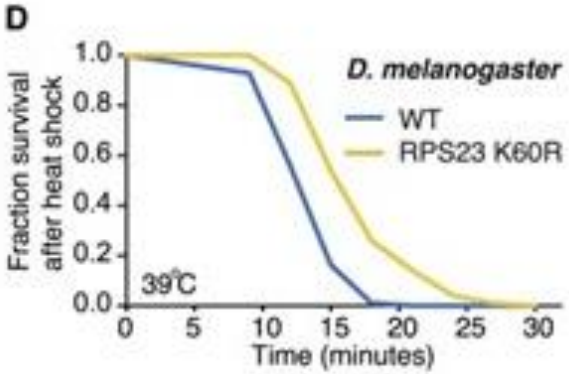
Protein mutations mutational rate and aging

蛋白质突变，突变率与衰老









In the process of translation, the frequency of translation mistakes is on the order of 10^{-3} to 10^{-6}

young and undamaged proteins altogether can further improve the longevity.

Cell Metabolism 33, 2288–2300, November 2, 2021

Seritonin, dopimin, cortison,….

- Vitamin B, ….
- AI medicine

Absence of final exam

- Here below is the information :
- Link for Exam Regulation for Student -
- https://registry.hkust.edu.hk/resource-library/examination-regulations-student
- Report on Illness or Other Circumstances Affecting Assessment :
- https://registry.hkust.edu.hk/system/files/internal/FormEX16.pdf
- This form should be completed by student and submit to course instructor.
- If any student may not be able to take final exam in person on campus, he/she needs to submit an official notification to ARO by the following steps:

1. Student needs to fill in ARO Form EX-16 Section I to Section III first.
2. Then, Course Instructor needs to complete Section IV.
3. Student submits the form with all the supporting document(s) to ARO Exam Team by email : arexamgrade@ust.hk.